



Chapter 2

Combinational logic gates

朱晓雷

xl_zhu@zju.edu.cn

2024

*Some Contents Adapted from Copyright Prentice Hall/Pearson , Rabaey's Digital Integrated Circuits, Mary Jane Irwin's VLSI Digital Circuits and University of Bolton, Weste's CMOS VLSI Design

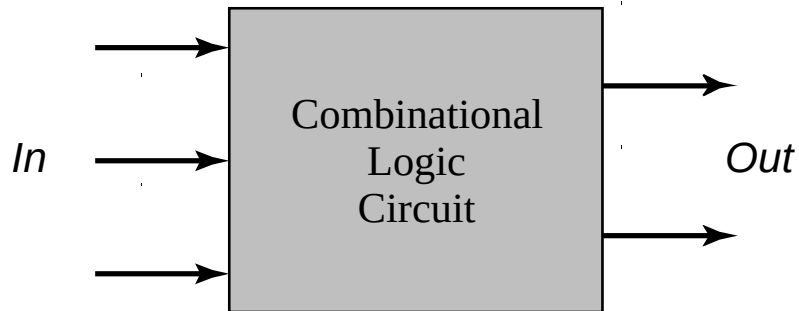


CMOS Circuit Styles

- **Static CMOS** - output connected to either V_{DD} or GND via a low-resistance path
 - high noise margins
 - low output impedance, high input impedance
 - no steady state path between V_{DD} and GND
 - delay a function of load capacitance and transistor resistance
- **Dynamic CMOS** - relies on temporary storage of signal values on the capacitance of high-impedance circuit nodes
 - simpler, faster gates
 - increased sensitivity to noise

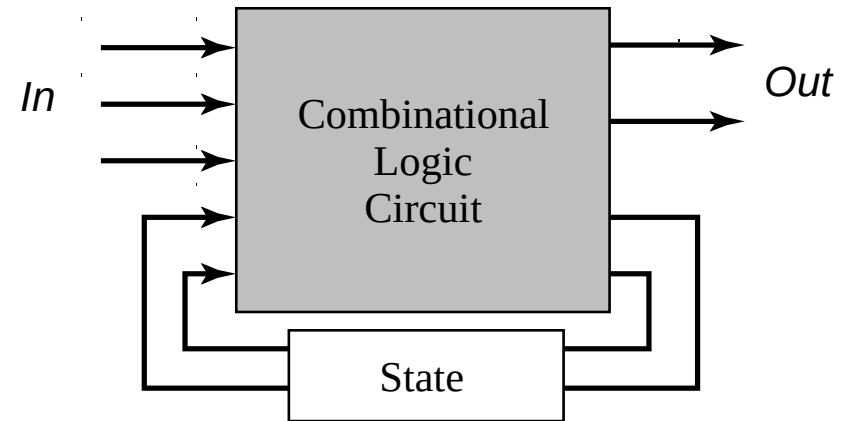


Combinational vs. Sequential Logic



Combinational

Output = $f(\text{In})$



Sequential

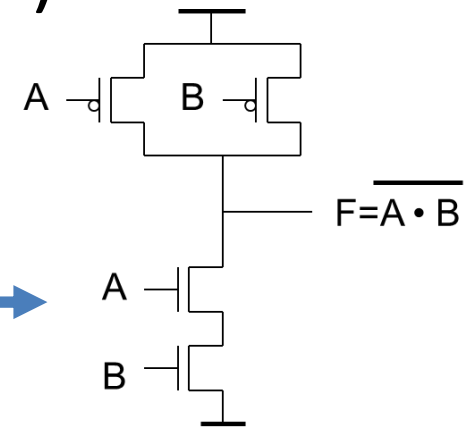
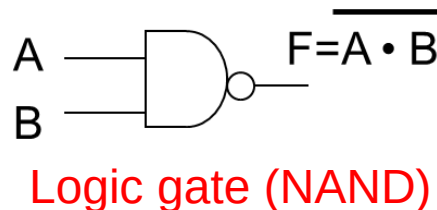
Output = $f(\text{In}, \text{Previous In})$



Combinational logic expressions

- Combinational logic: function value is a combination of function arguments.
- A logic gate implements a particular logic function.
- Both specification (logic equations) and implementation (logic gate networks) are written in Boolean logic.

$F = A \cdot B$
Logic function

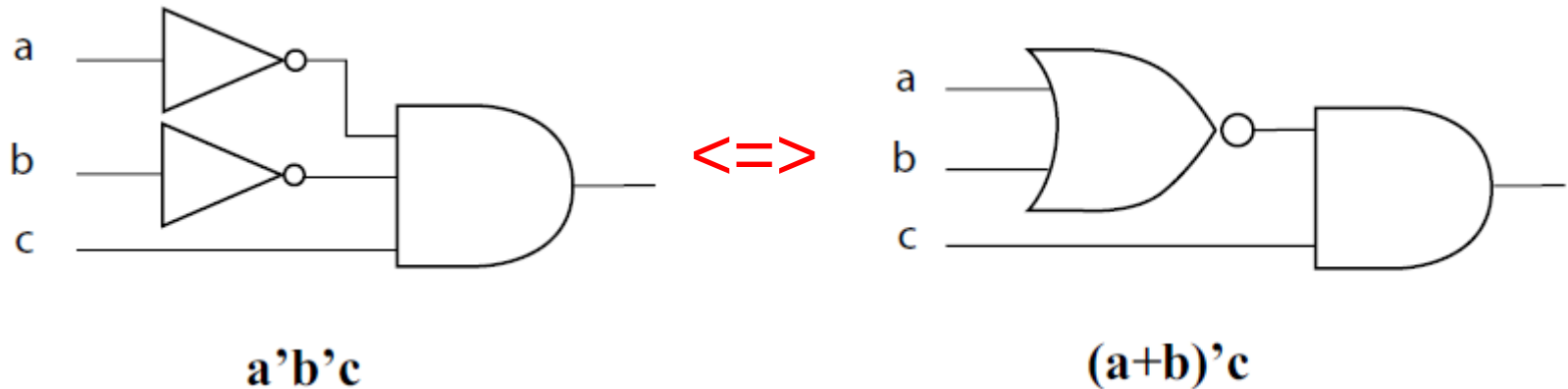




Gate design

Why designing gates for logic functions is important:

- may not have logic gates in the library for all logic expressions;
- a logic expression may map into gates that consume a lot of area, delay, or power.



$$(a + b)'c = a'b'c$$



Boolean algebra terminology

- Function:

- $f = a'b + ab'$

- a is a variable; a and a' are literal variables.

- ab' is a term.

- A function is irredundant if no literal can be removed without changing its truth value.



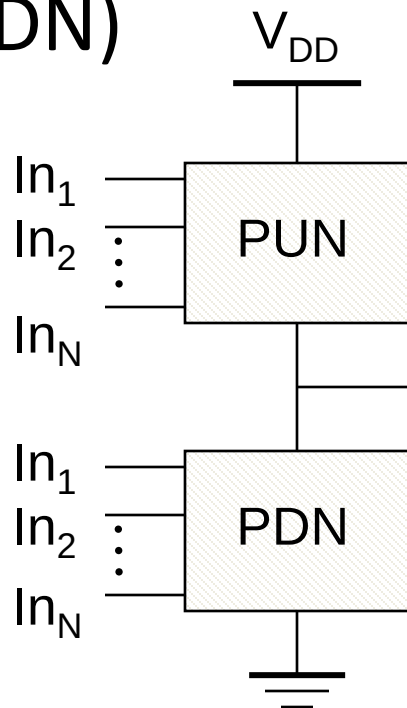
Completeness

- A set of functions $f_1, f_2, \dots$ is complete if every Boolean function can be generated by a combination of the functions.
- NAND is a complete set; NOR is a complete set; {AND, OR} is not complete.
- If your set of logic gates is not complete, you can't design arbitrary logic. $ab + ab' = a$
- Simplifying logic expressions before designing the gate network is important for both area and delay.



Static Complementary CMOS

- Pull-up network (PUN) and pull-down network (PDN)



PMOS transistors only

pull-up: make a connection from V_{DD} to F when $F(In_1, In_2, \dots, In_N) = 1$

pull-down: make a connection from F to GND when $F(In_1, In_2, \dots, In_N) = 0$

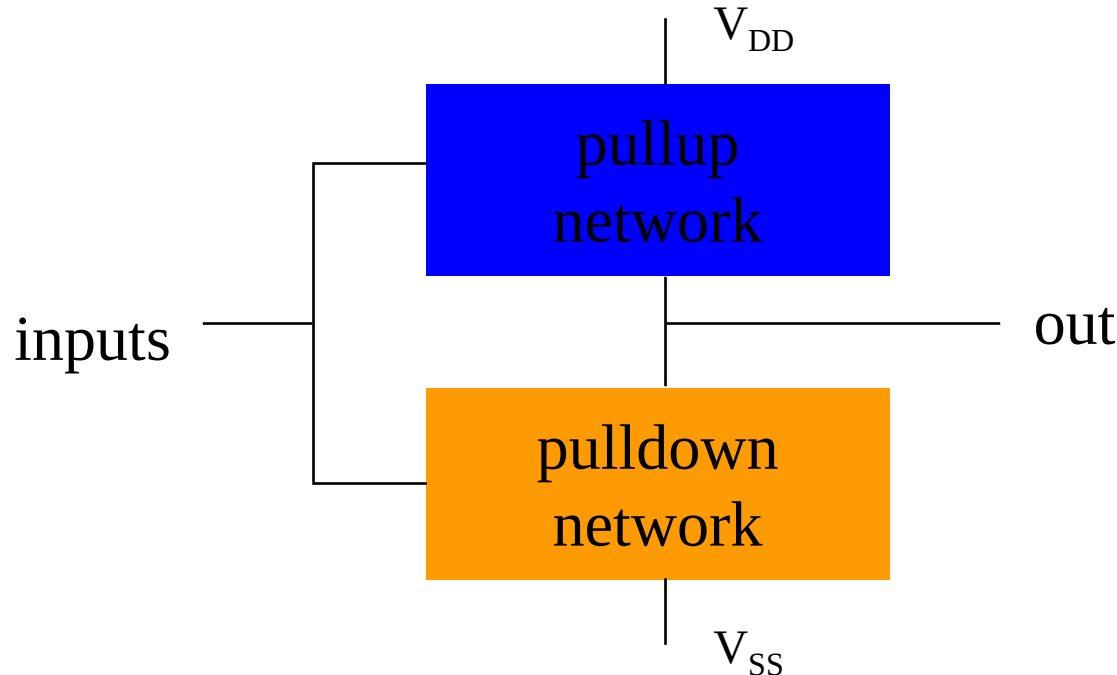
NMOS transistors only

PUN and PDN are dual logic networks



Static complementary gates

- Complementary: have complementary pullup (p-type) and pulldown (n-type) networks.
- Static: do not rely on stored charge.
- Simple, effective, reliable; hence ubiquitous.

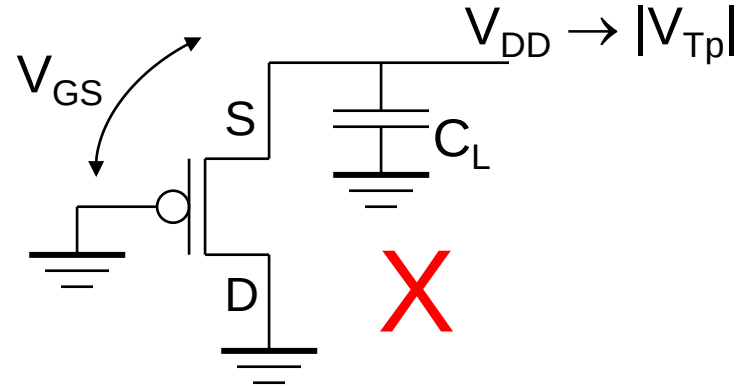
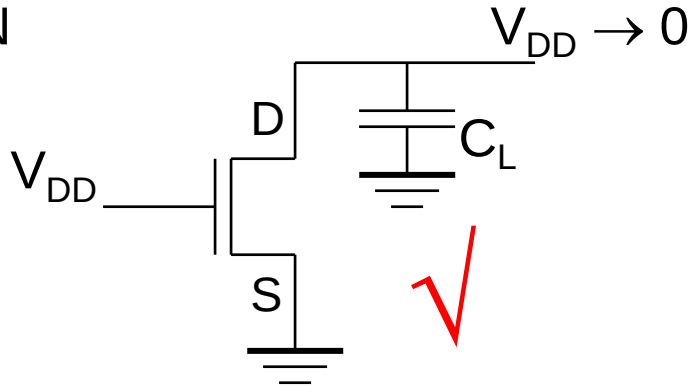




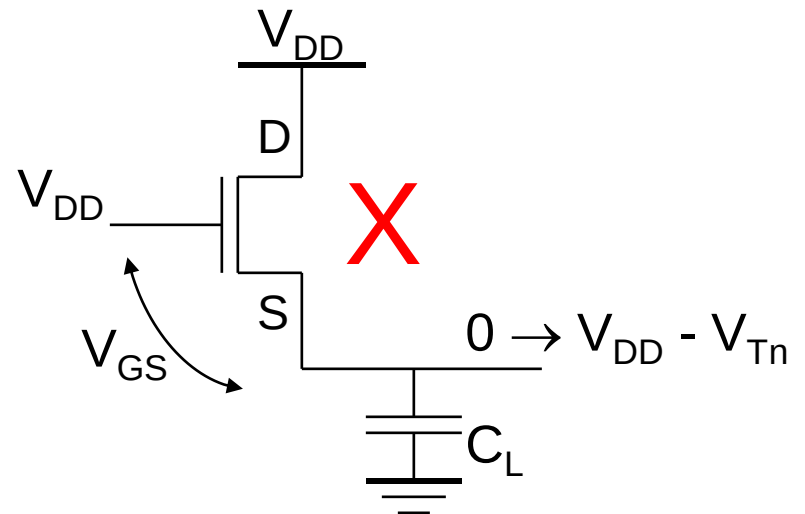
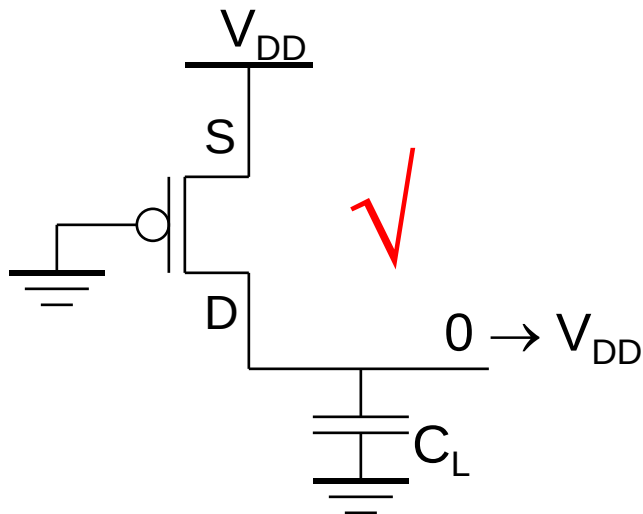
Transistor types selected for PDN & PUN

Why PMOS in PUN and NMOS in PDN

PDN

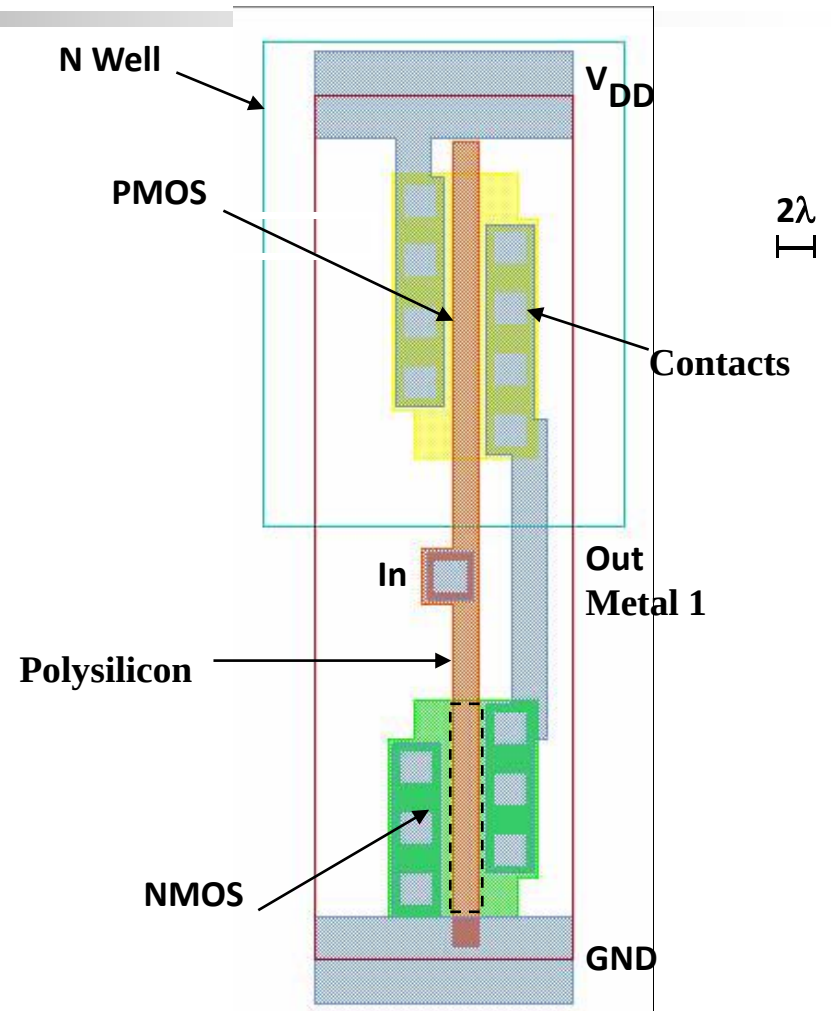
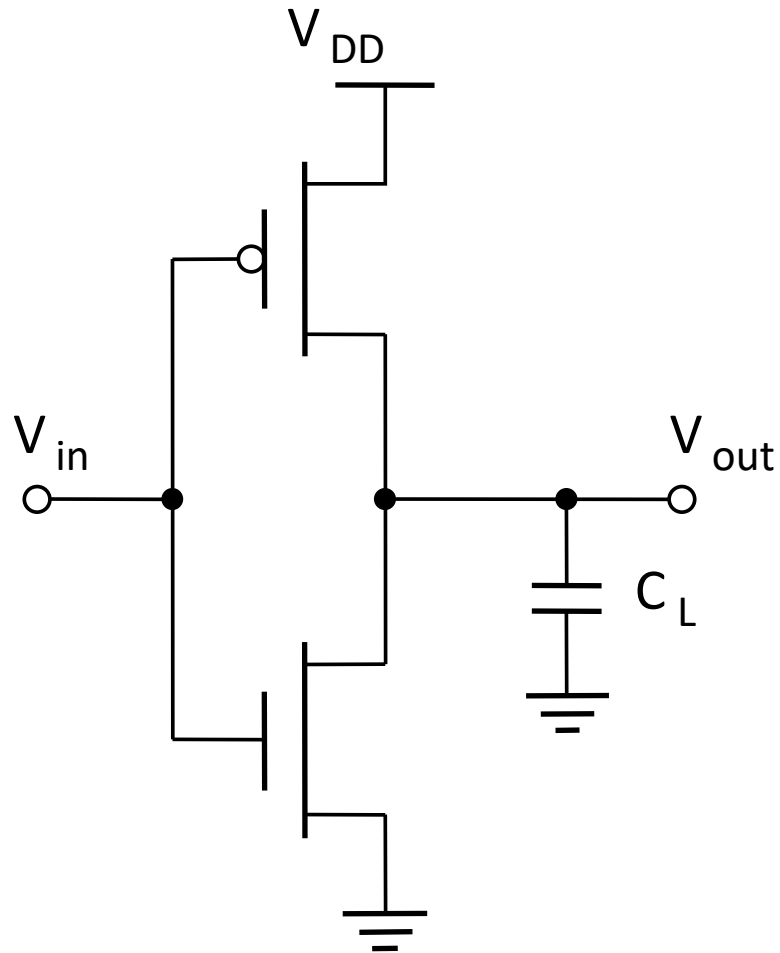


PUN



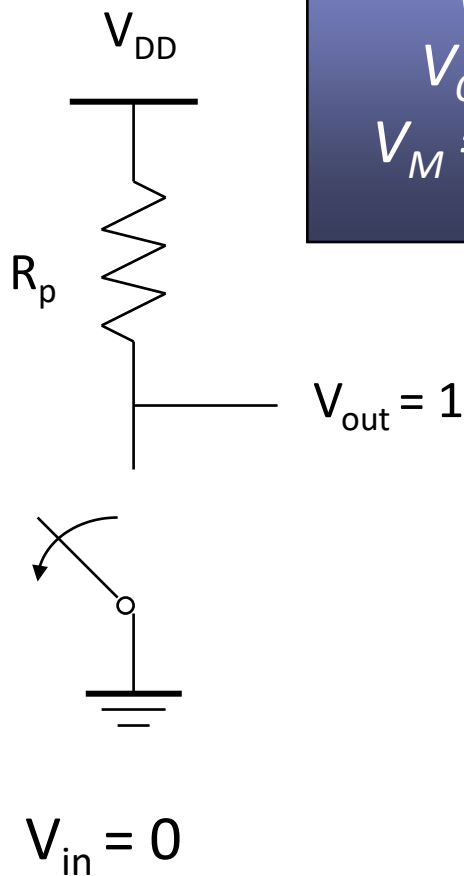
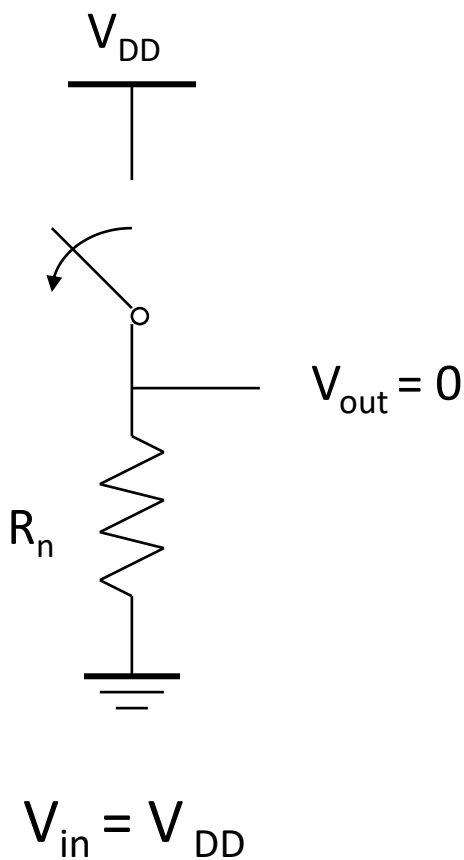


The CMOS Inverter

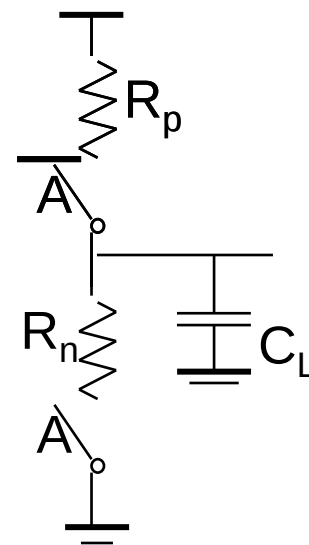




First-Order DC Analysis



$$\begin{aligned} V_{OL} &= 0 \\ V_{OH} &= V_{DD} \\ V_M &= f(R_n, R_p) \end{aligned}$$



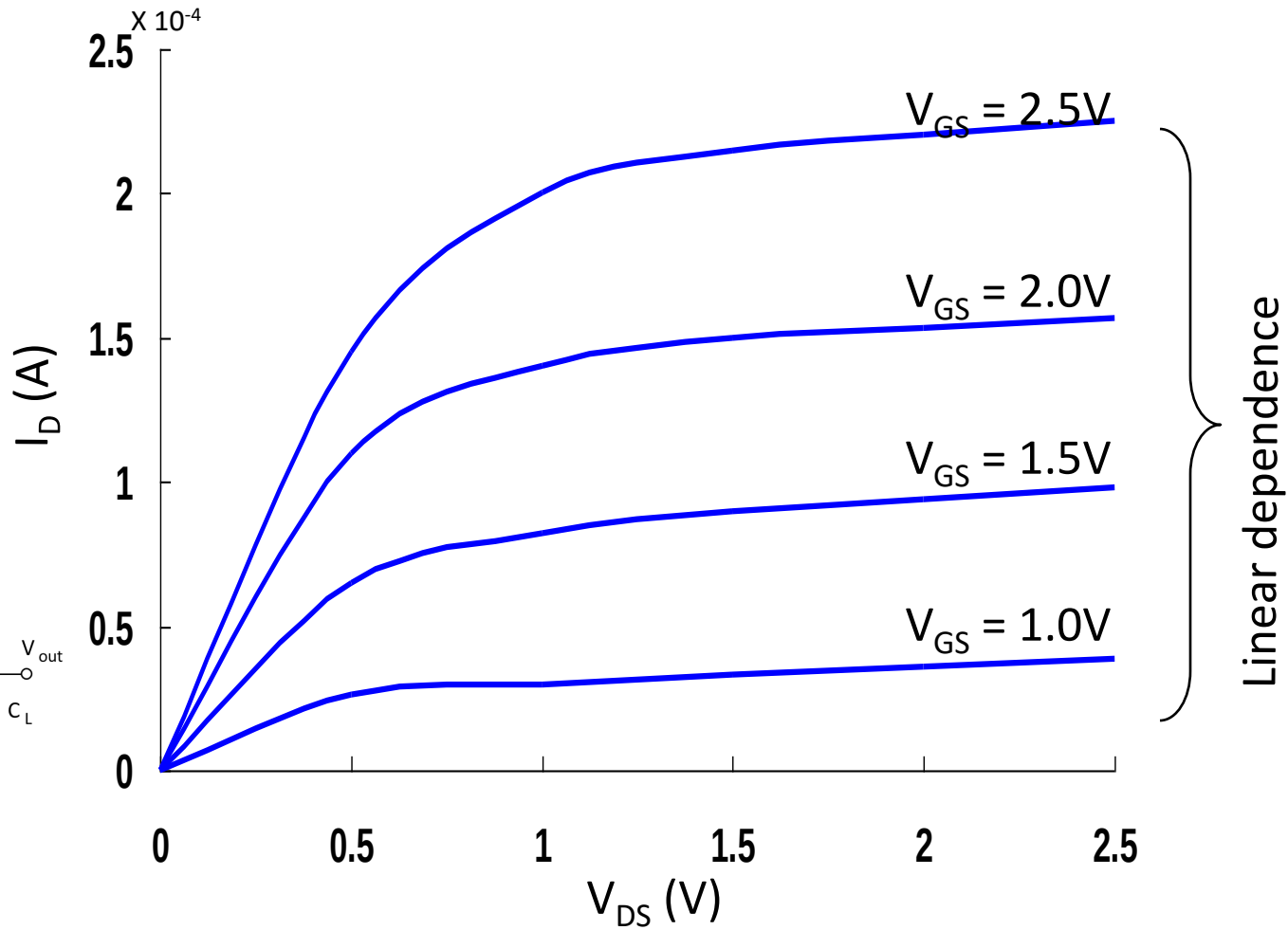


Properties of the Inverter

- Full rail-to-rail swing $\Rightarrow$ **high noise margins**
 - Logic levels not dependent upon the relative device sizes $\Rightarrow$ transistors can be minimum size $\Rightarrow$ **ratioless**
- Always a path to V_{dd} or GND in steady state $\Rightarrow$ **low output impedance** (output resistance in $k\Omega$ range) $\Rightarrow$ **large fan-out** (albeit with degraded performance)
- **Extremely high input resistance** (gate of MOS transistor is near perfect insulator) $\Rightarrow$ nearly zero steady-state input current
- No direct path steady-state between power and ground $\Rightarrow$ **no static power dissipation** (ignores leakage current)
- Propagation delay function of load capacitance and resistance of transistors



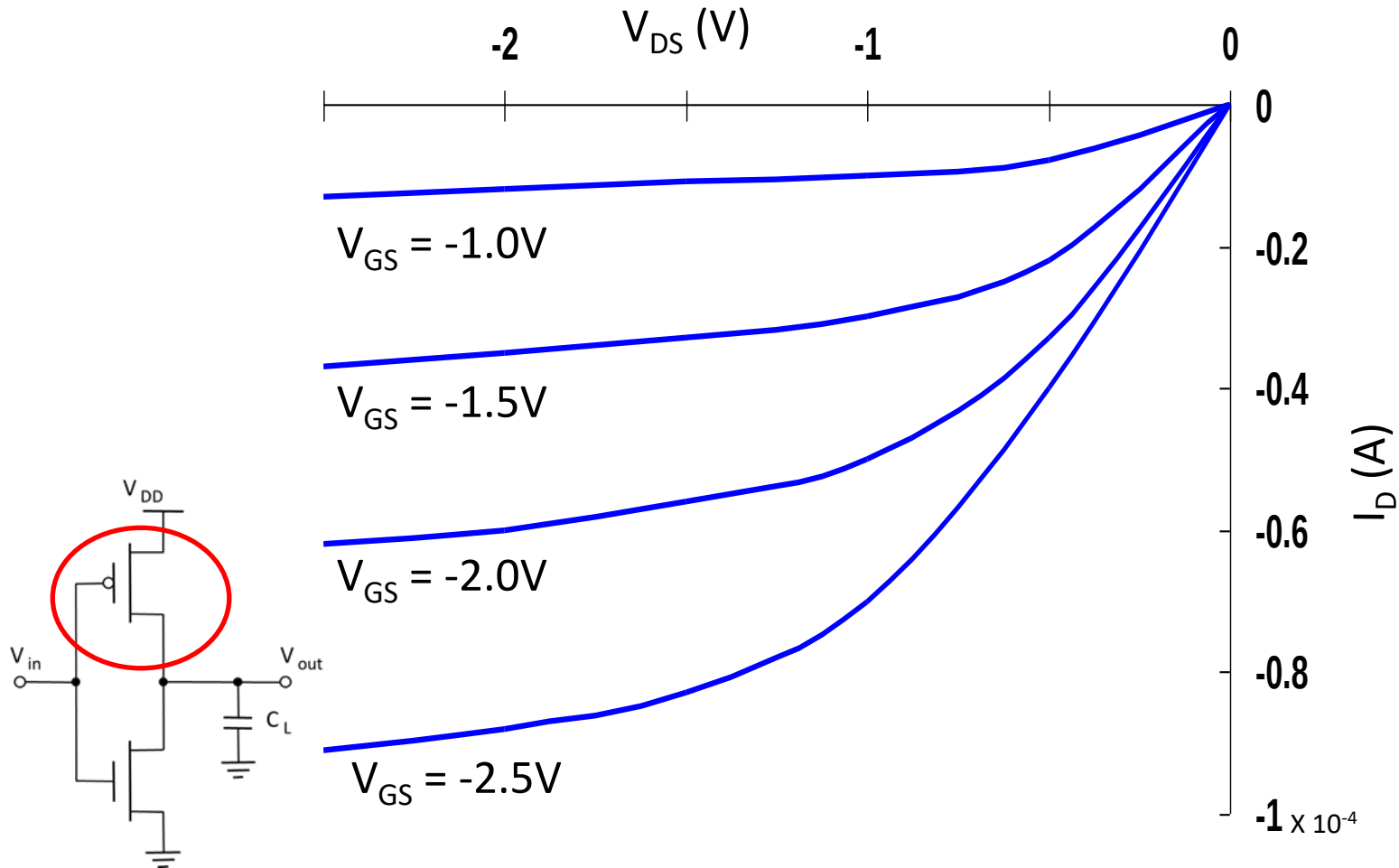
Review: Short Channel I-V Plot (NMOS)



NMOS transistor, $0.25\mu m$, $L_d = 0.25\mu m$, $W/L = 1.5$, $V_{DD} = 2.5V$, $V_T = 0.4V$
Digital IC Design



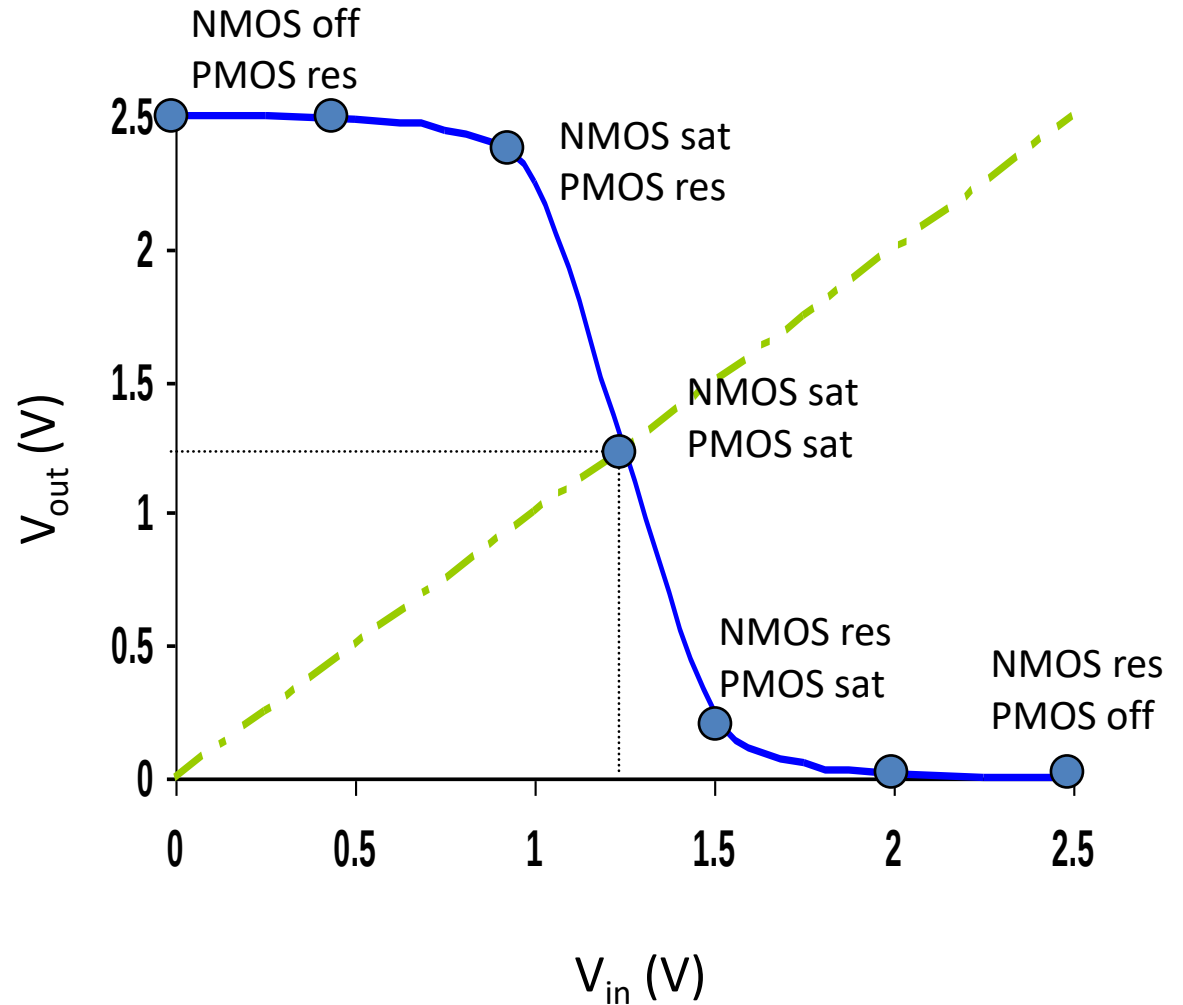
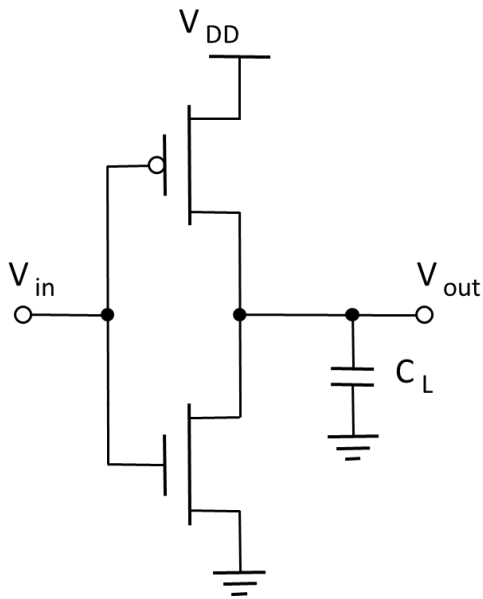
Review: Short Channel I-V Plot (PMOS)



PMOS transistor, $0.25\mu\text{m}$, $L_d = 0.25\mu\text{m}$, $W/L = 1.5$, $V_{DD} = 2.5V$, $V_T = -0.4V$

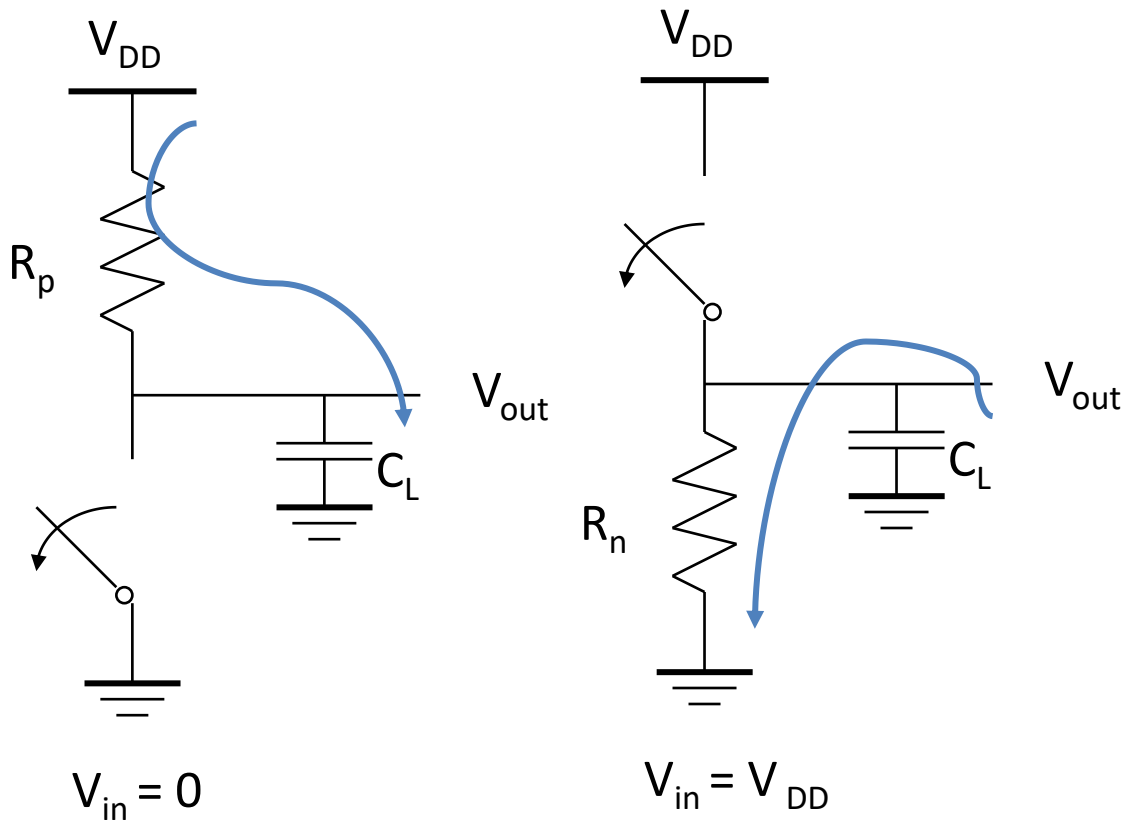


CMOS Inverter VTC





Switch Model of Dynamic Behavior



Gate response time is determined by the time to charge C_L through R_p
(discharge C_L through R_n)

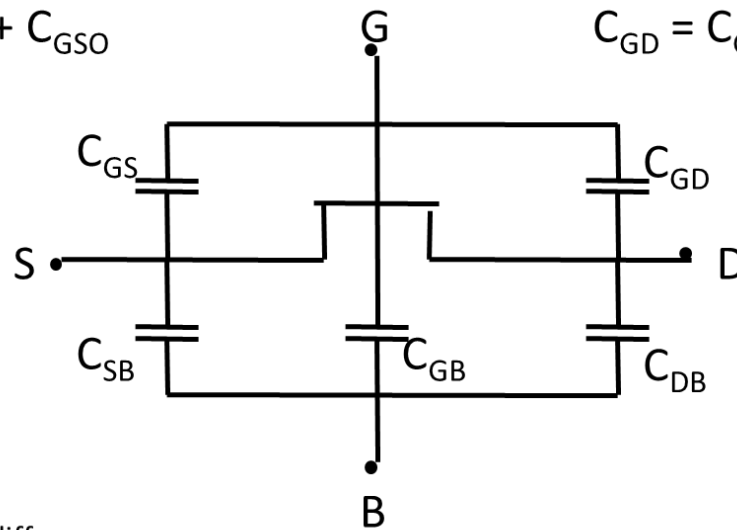


Revisit Parasitics MOS Capacitance

- Gate capacitance
 - Gate-overlap capacitance
 - Gate-channel capacitance
- Diffusion capacitance

$$C_{GS} = C_{GCS} + C_{GSO}$$

$$C_{GD} = C_{GCD} + C_{GDO}$$



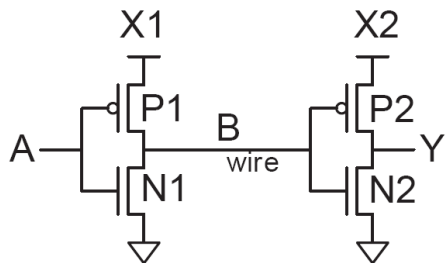
$$C_{SB} = C_{Sdiff}$$

$$C_{DB} = C_{Ddiff}$$

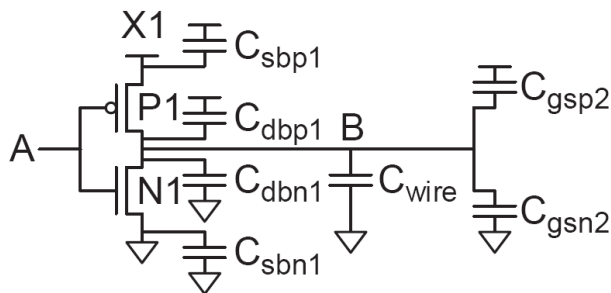
$$C_{GB} = C_{GCB}$$



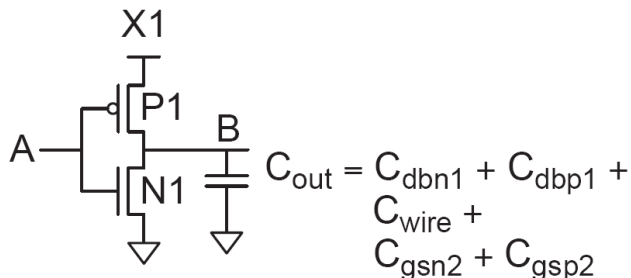
Summary on Inverter Capacitance



(a)



(b)



(c)

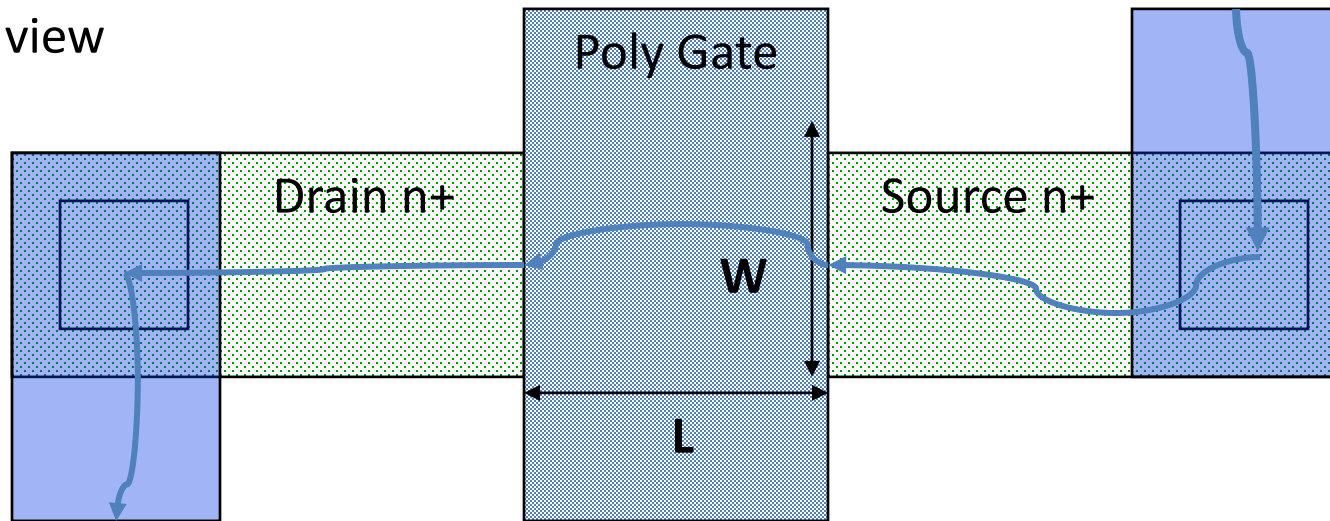
- Capacitance is proportional to $W \cdot L$ of the transistor
- Cell typically in the order of fF ($10^{-15}F$)
- The total C_{die} of a chip may go to hundreds to thousands of nF or more
 - Back-calculate how many transistors

FIGURE 4.3 Capacitances for inverter delay calculations



Sources of Resistance

Top view



- MOS structure resistance - $R_{on} \propto R_{n0/p0} L/W$
- Source and drain resistance
- Contact (via) resistance
- Wiring resistance

$$R_{p0}/R_{n0}=2\sim4$$

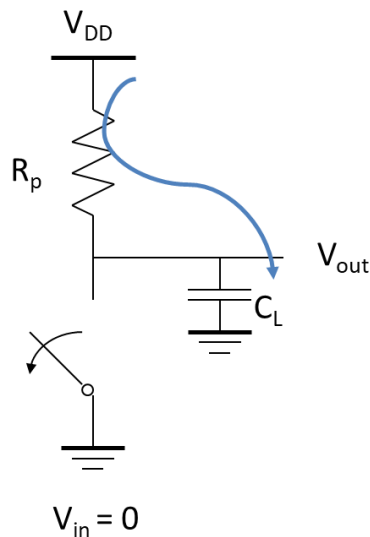


For equal R_{on} of NMOS and PMOS
 $W_p/W_n=2\sim4$

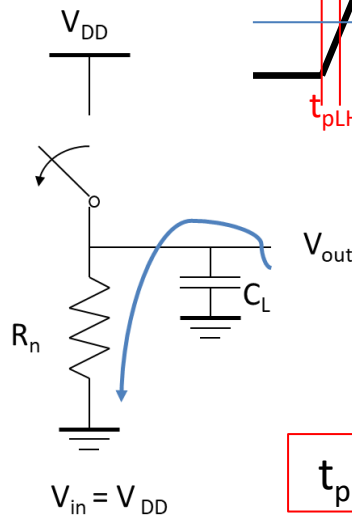


Inverter Propagation Delay

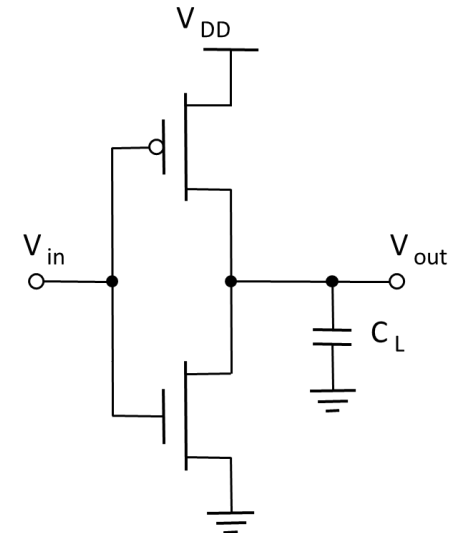
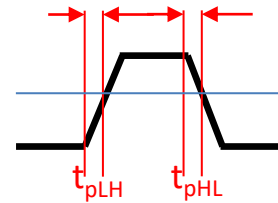
- Propagation delay is proportional to the time-constant of the network formed by the pull-down resistor and the load capacitance. $t_{pHL} = f(R_n, C_L)$



$$t_{pLH} = \ln(2) R_p C_L = 0.69 R_p C_L$$



$$t_{pHL} = \ln(2) R_n C_L = 0.69 R_n C_L$$



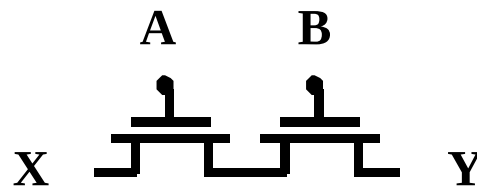
$$t_p = (t_{pHL} + t_{pLH})/2 = 0.69 C_L (R_n + R_p)/2$$

- To equalize rise and fall times make the on-resistance of the NMOS and PMOS approximately equal.

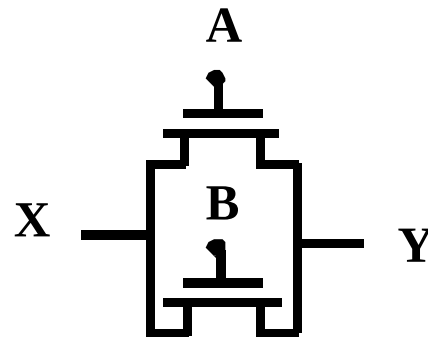


NMOS Transistors in Series/Parallel Connection

- Transistors can be thought as a switch controlled by its gate signal
- NMOS switch closes when switch control input is high



$Y = X$ if A and B

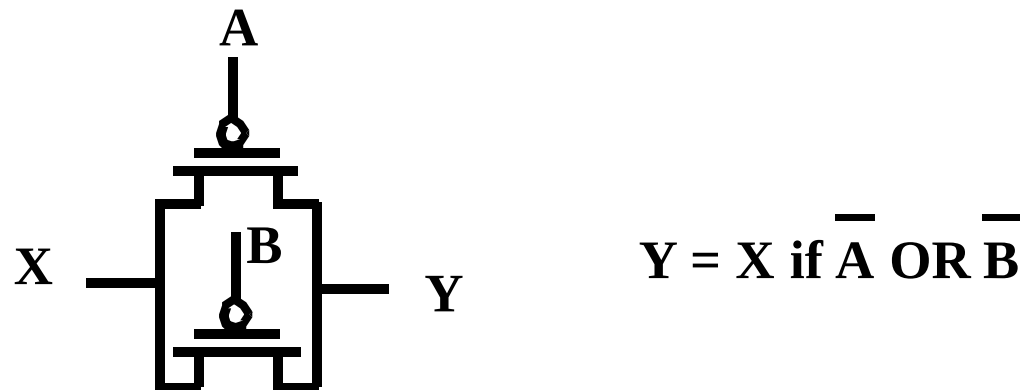
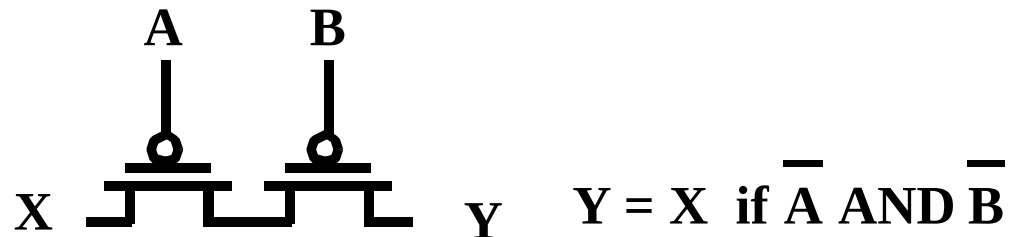


$Y = X$ if A OR B



PMOS Transistors in Series/Parallel Connection

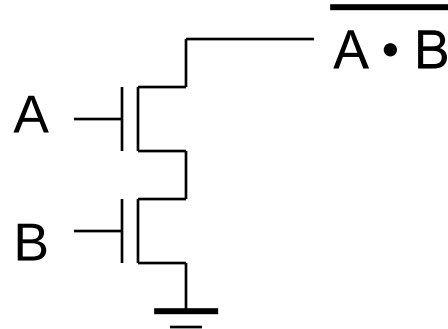
- PMOS switch closes when switch control input is low



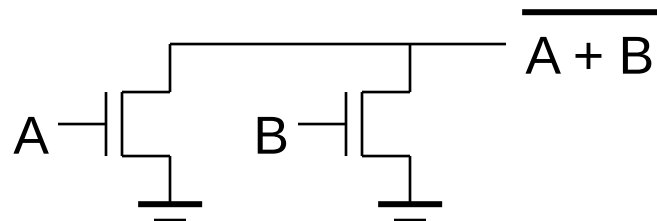


Construction of PDN

- NMOS devices in **series** implement a NAND function



- NMOS devices in **parallel** implement a NOR function





Dual PUN and PDN

- PUN and PDN are dual networks

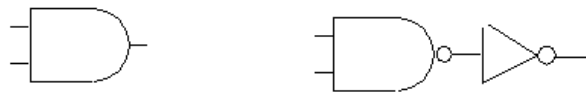
- De Morgan's theorems

$$\overline{A + B} = \overline{A} \cdot \overline{B} \quad [!(A + B) = !A \cdot !B \text{ or } !(A \mid B) = !A \& !B]$$

$$\overline{A \cdot B} = \overline{A} + \overline{B} \quad [!(A \cdot B) = !A + !B \text{ or } !(A \& B) = !A \mid !B]$$

- a **parallel** connection of transistors in the PUN corresponds to a **series** connection of the PDN

- Complementary gate is naturally **inverting** (NAND, NOR)

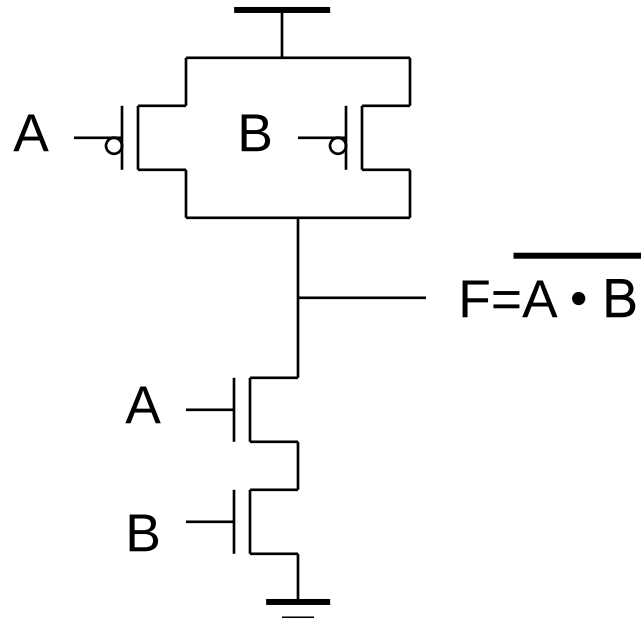


$$\text{AND} = \text{NAND} + \text{INV}$$

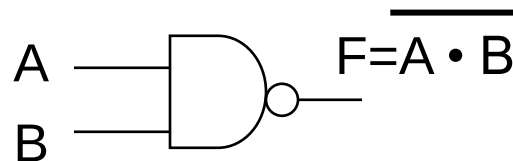
- Number of transistors for N-input logic gate: **2N**



Example: CMOS NAND

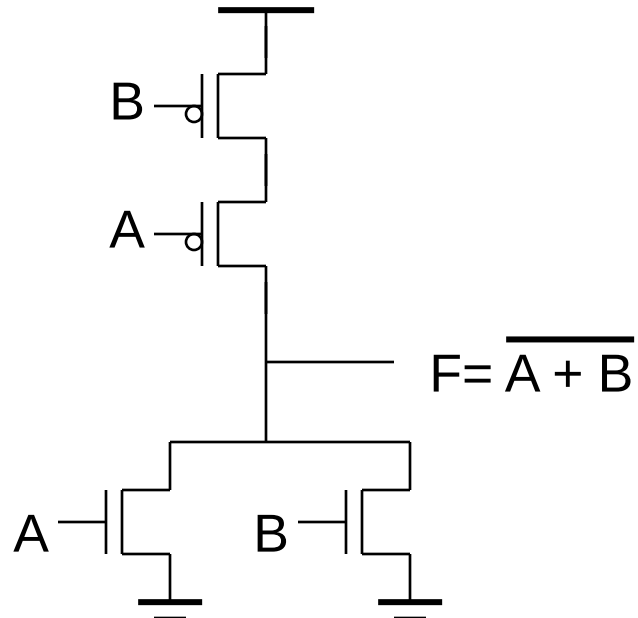


A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

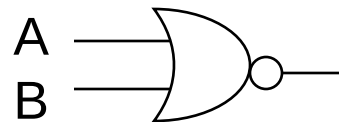




CMOS NOR



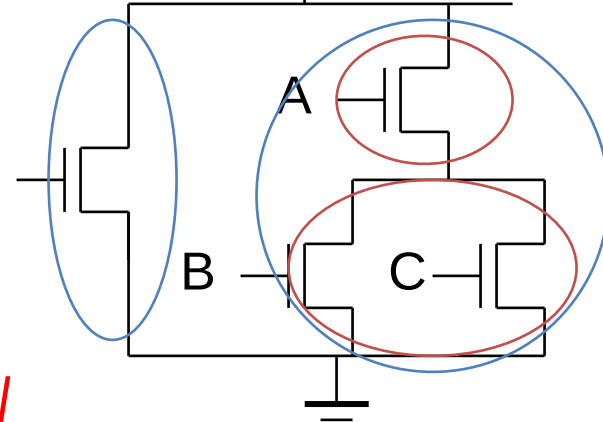
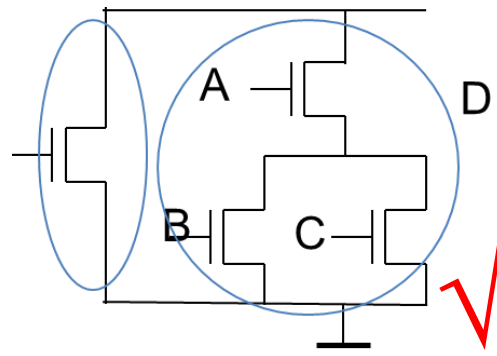
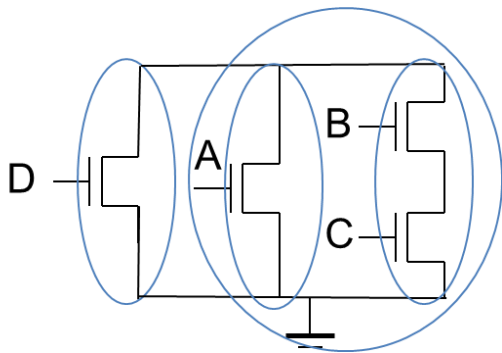
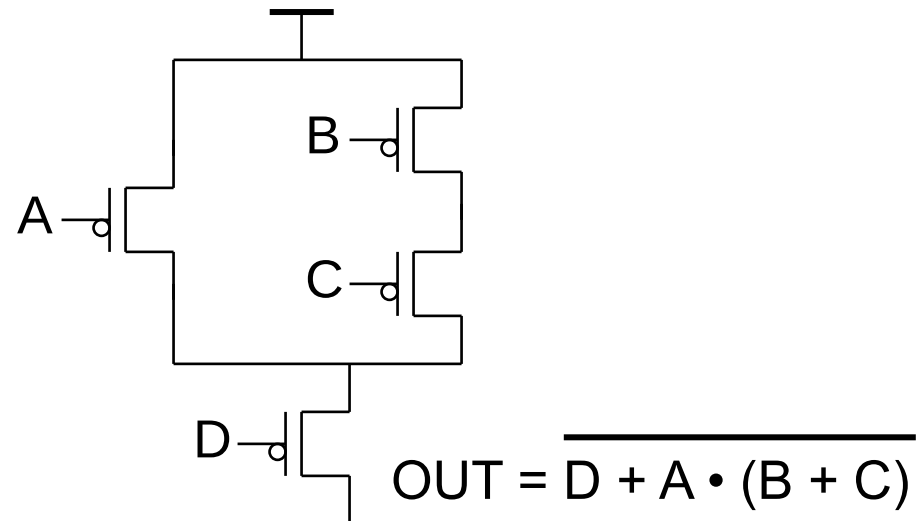
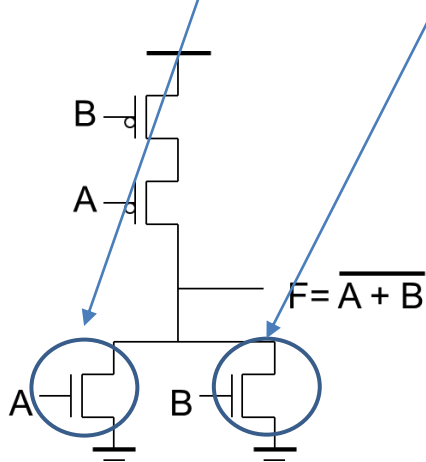
A	B	F
0	0	1
0	1	0
1	0	0
1	1	0





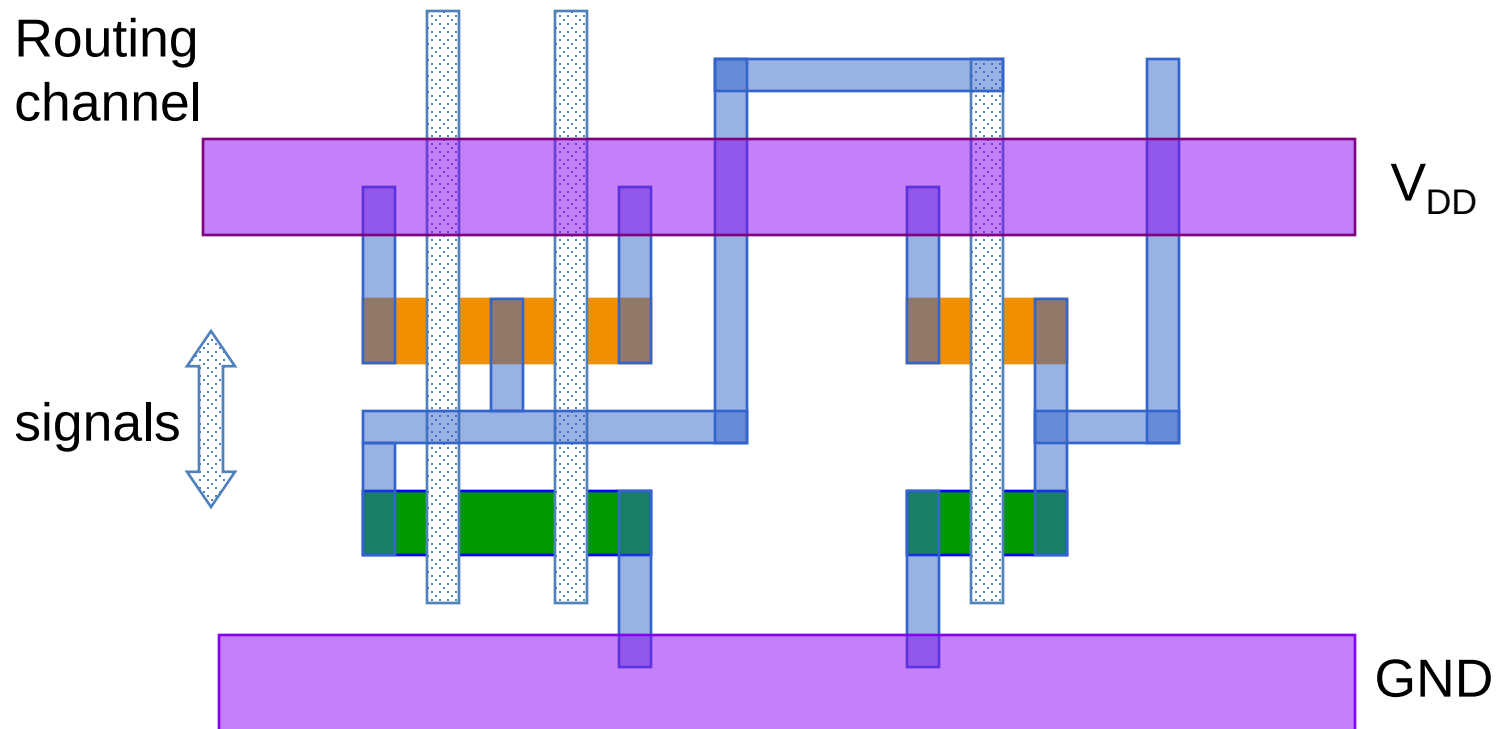
Complex CMOS Gate

$$\text{OUT} = \overline{D + A \cdot (B + C)}$$





Cell Layout



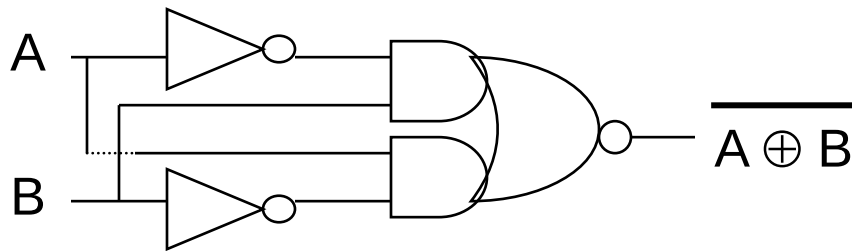
What logic function is this?

$$Y=AB$$

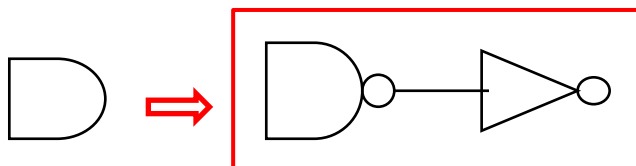
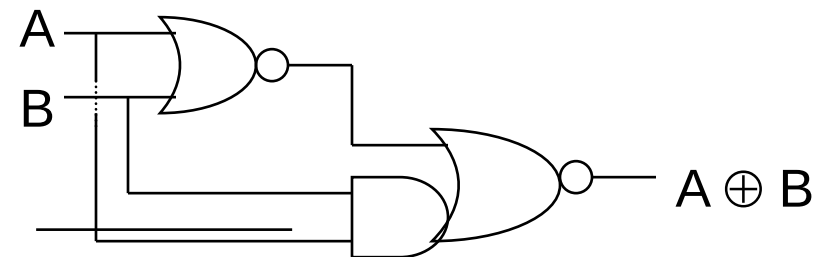
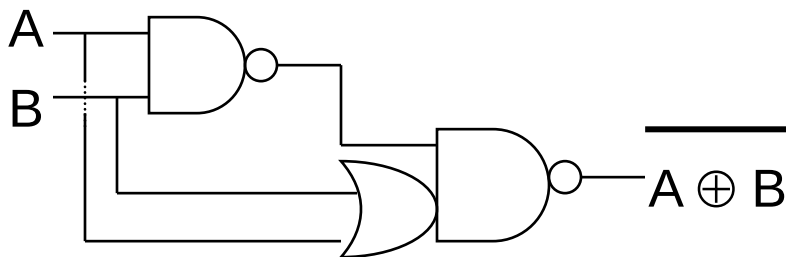
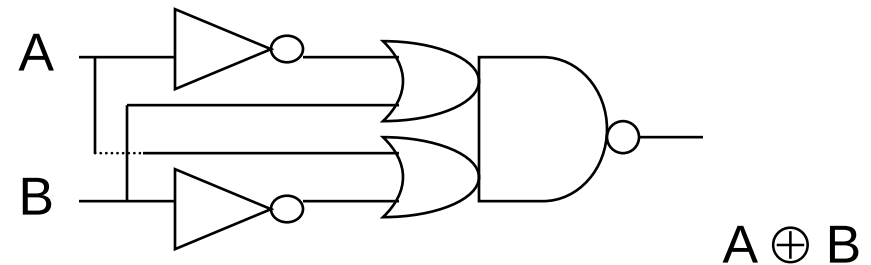


XNOR/XOR Implementation

$$\text{XNOR} = (A'B + AB')'$$

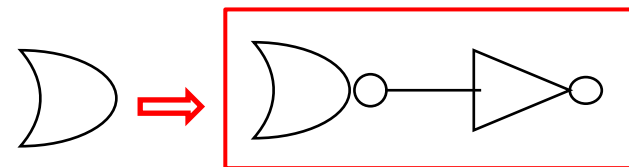


$$\text{XOR} = A'B + AB'$$



6 transistors

Digital IC Design



6 transistors



AOI/OAI gates

- AOI = and/or/invert; OAI = or/and/invert.
- Implement larger functions.
- Pullup and pulldown networks are compact: smaller area, higher speed than NAND/NOR network equivalents.



AOI example

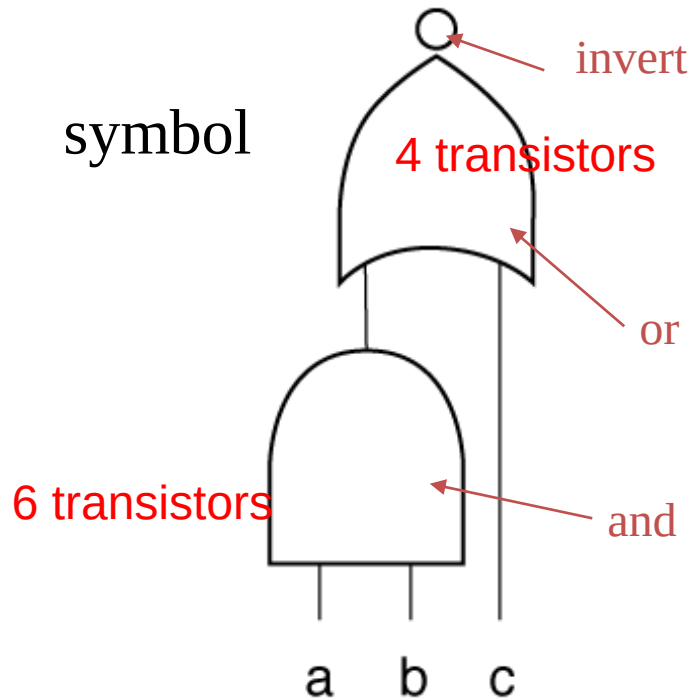
out = $[ab+c]'$:

Example

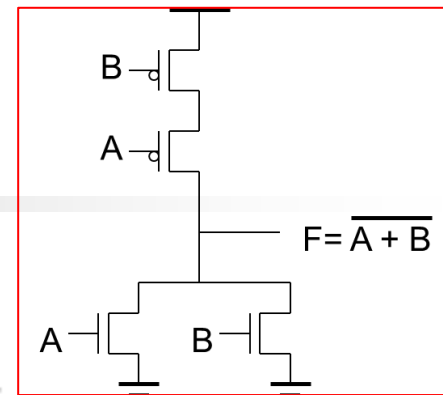
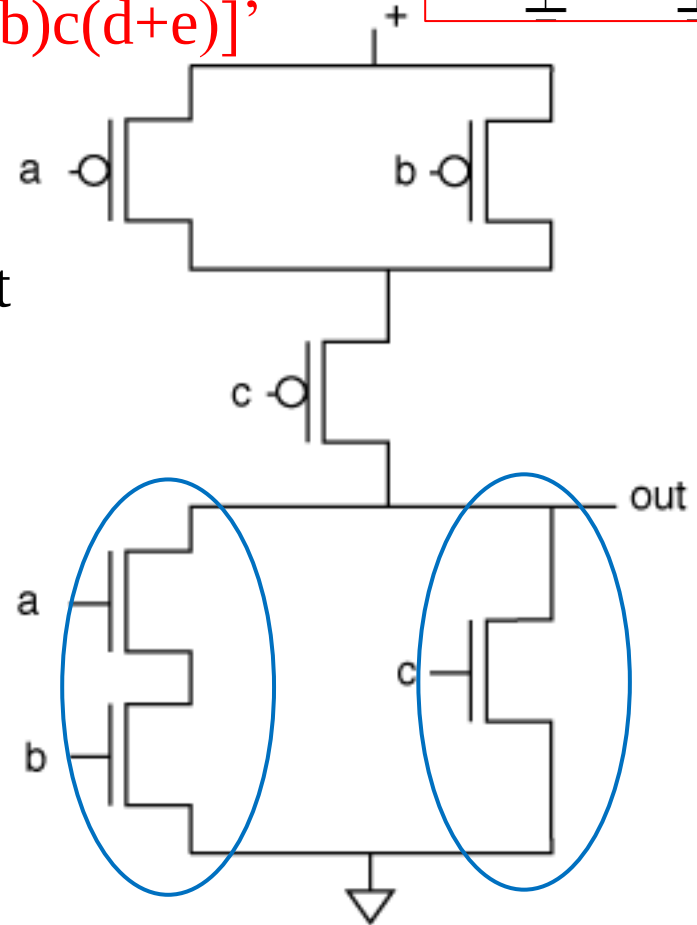
AOI-21: $(ab+c)'$

OAI-212: $[(a+b)c(d+e)]'$

symbol



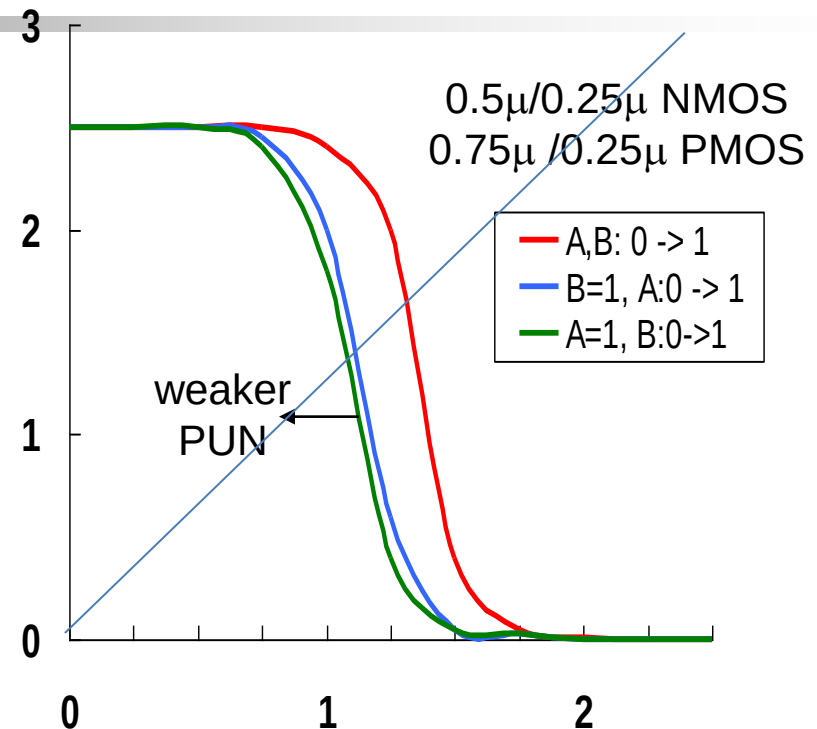
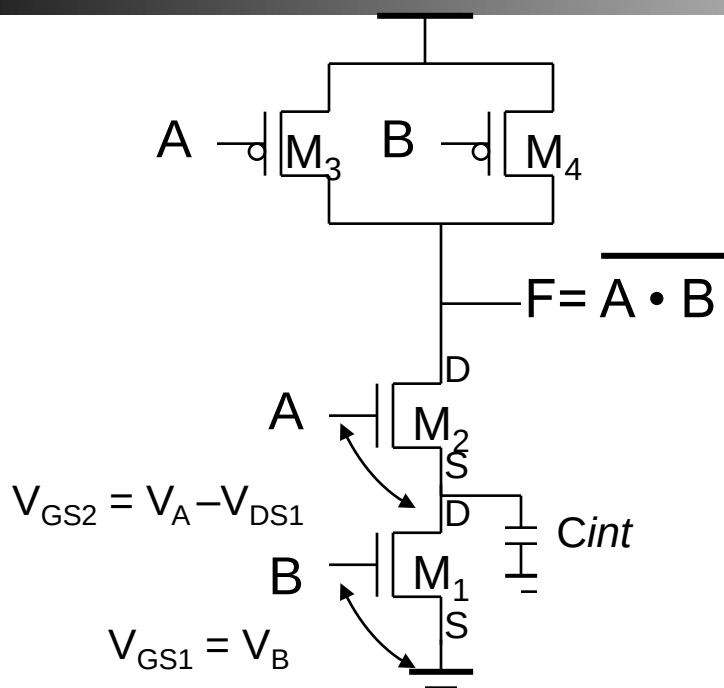
circuit



AOI-21



VTC is Data-Dependent



- The threshold voltage of M_2 is higher than M_1 due to the body effect (γ)

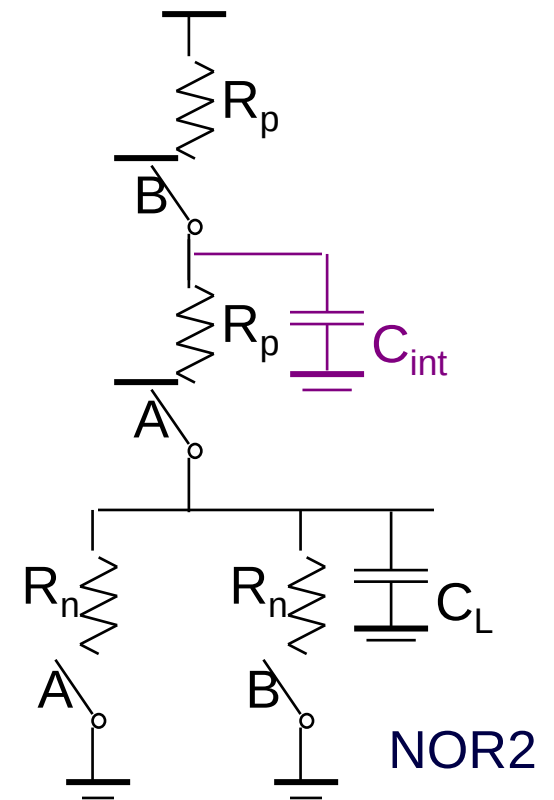
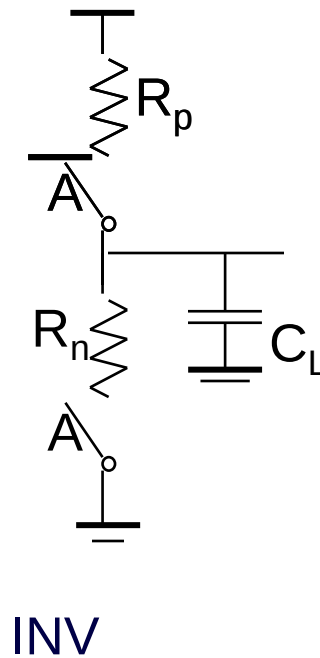
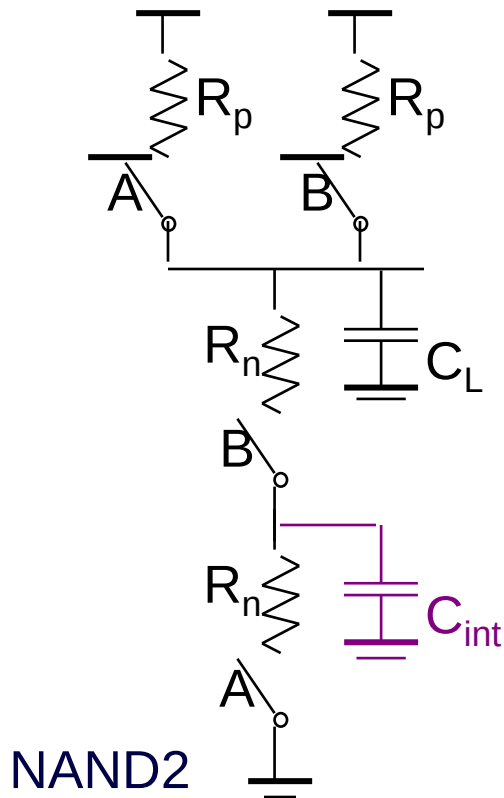
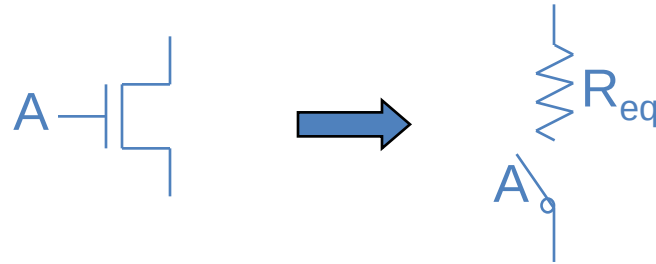
$$V_{Tn1} = V_{Tn0}$$

$$V_{Tn2} = V_{Tn0} + \gamma(\sqrt{(|2\phi_F| + V_{int})} - \sqrt{|2\phi_F|})$$

since V_{SB} of M_2 is not zero (when $V_B = 0$) due to the presence of C_{int}

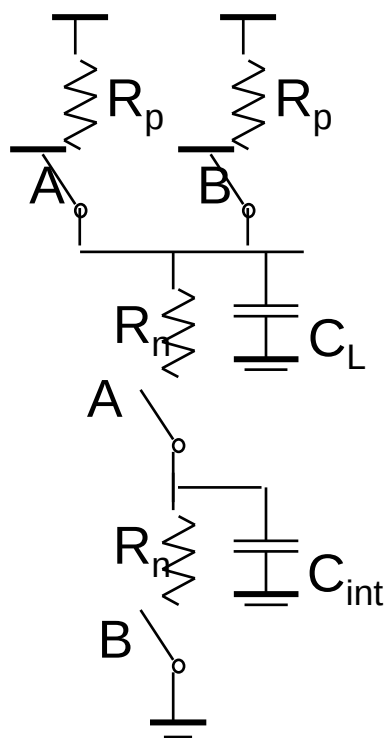


Switch Delay Model





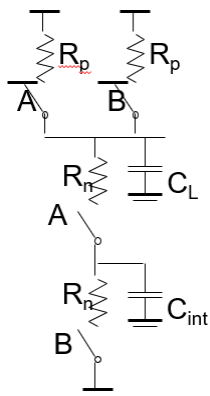
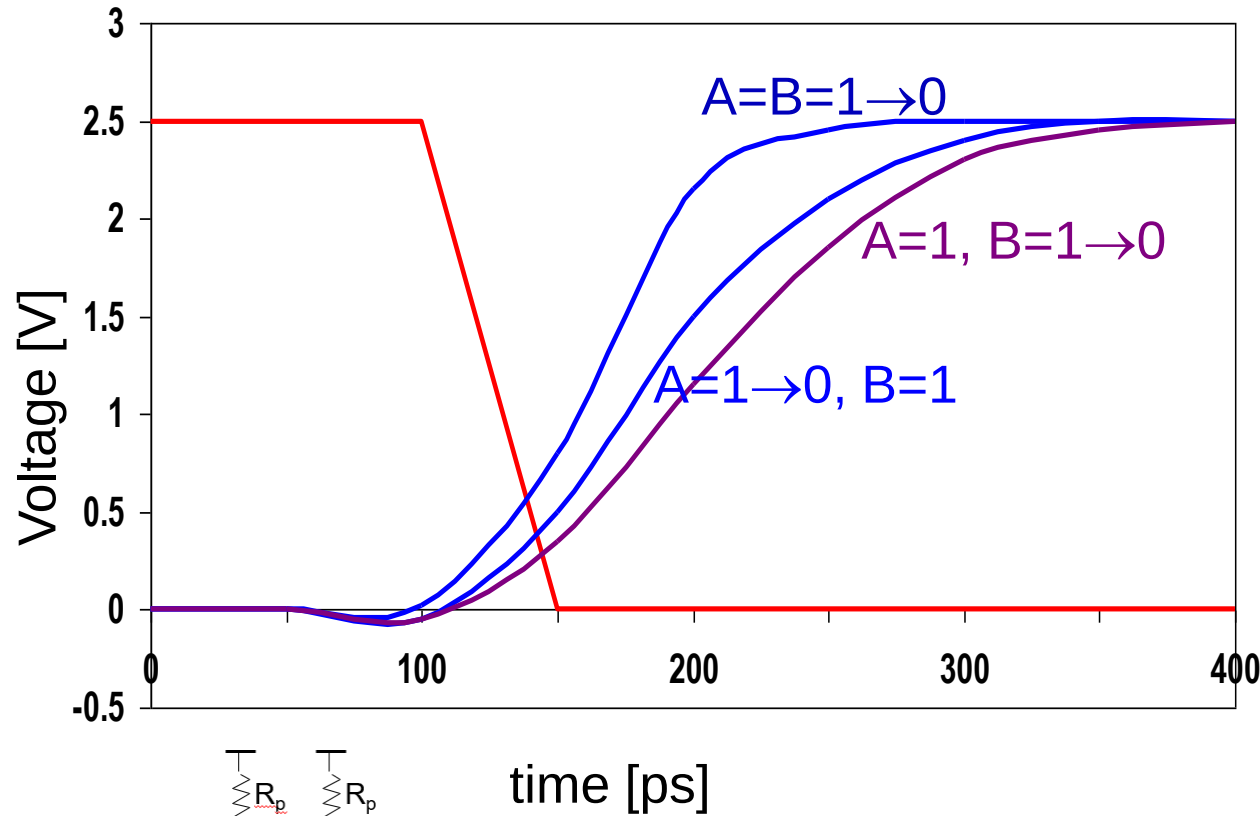
Input Pattern Effects on Delay



- Delay is dependent on the **pattern** of inputs
- Low to high transition at out put
 - both inputs go low
 - delay is $0.69 R_p/2 C_L$
 - one input goes low
 - delay is $0.69 R_p C_L$
- High to low transition
 - both inputs go high
 - delay is $0.69 2R_n C_L$



Delay Dependence on Input Patterns

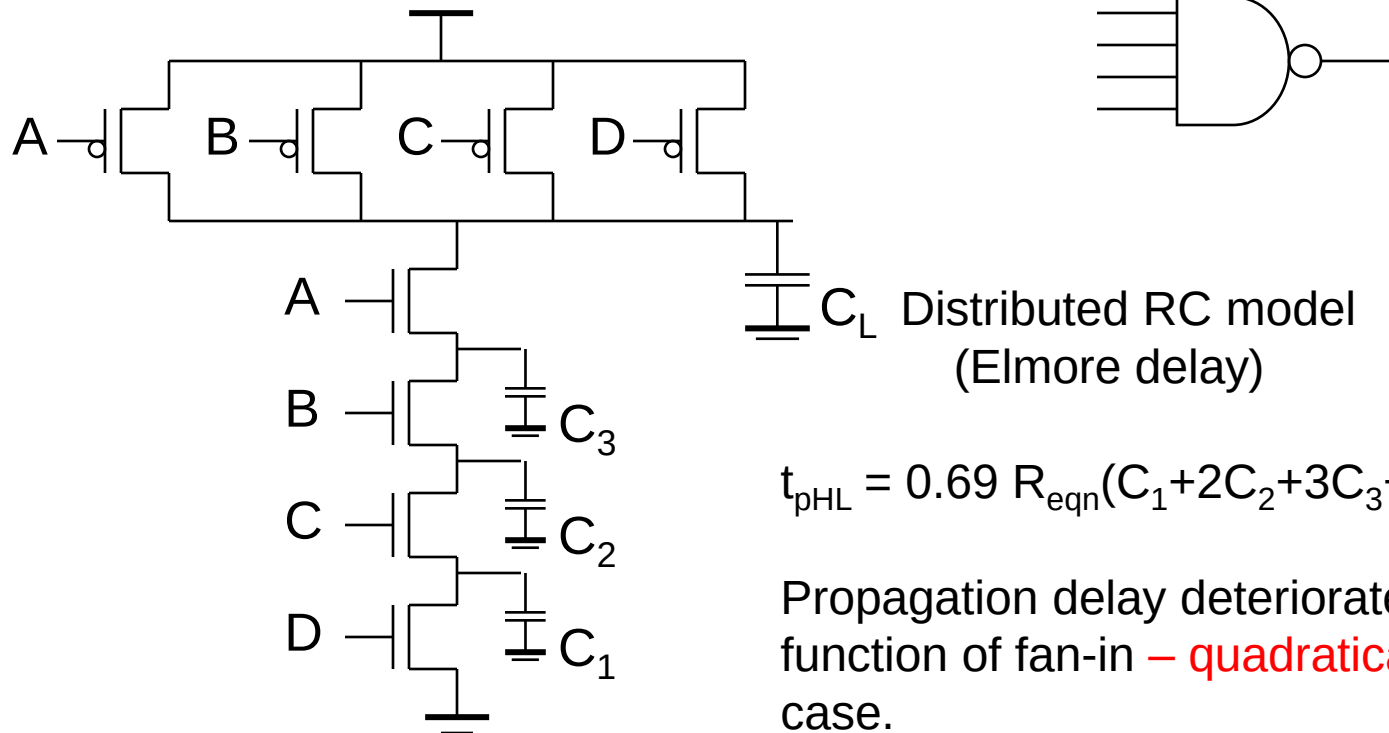


Input Data Pattern	Delay (psec)
$A=B=0 \rightarrow 1$	67
$B=1, A=0 \rightarrow 1$	64
$A=1, B=0 \rightarrow 1$	61
$A=B=1 \rightarrow 0$	45
$A=1, B=1 \rightarrow 0$	81
$A=1 \rightarrow 0, B=1$	80

NMOS = $0.5\mu\text{m}/0.25\mu\text{m}$
 PMOS = $0.75\mu\text{m}/0.25\mu\text{m}$
 $C_L = 100\text{ fF}$



Fan-In Considerations

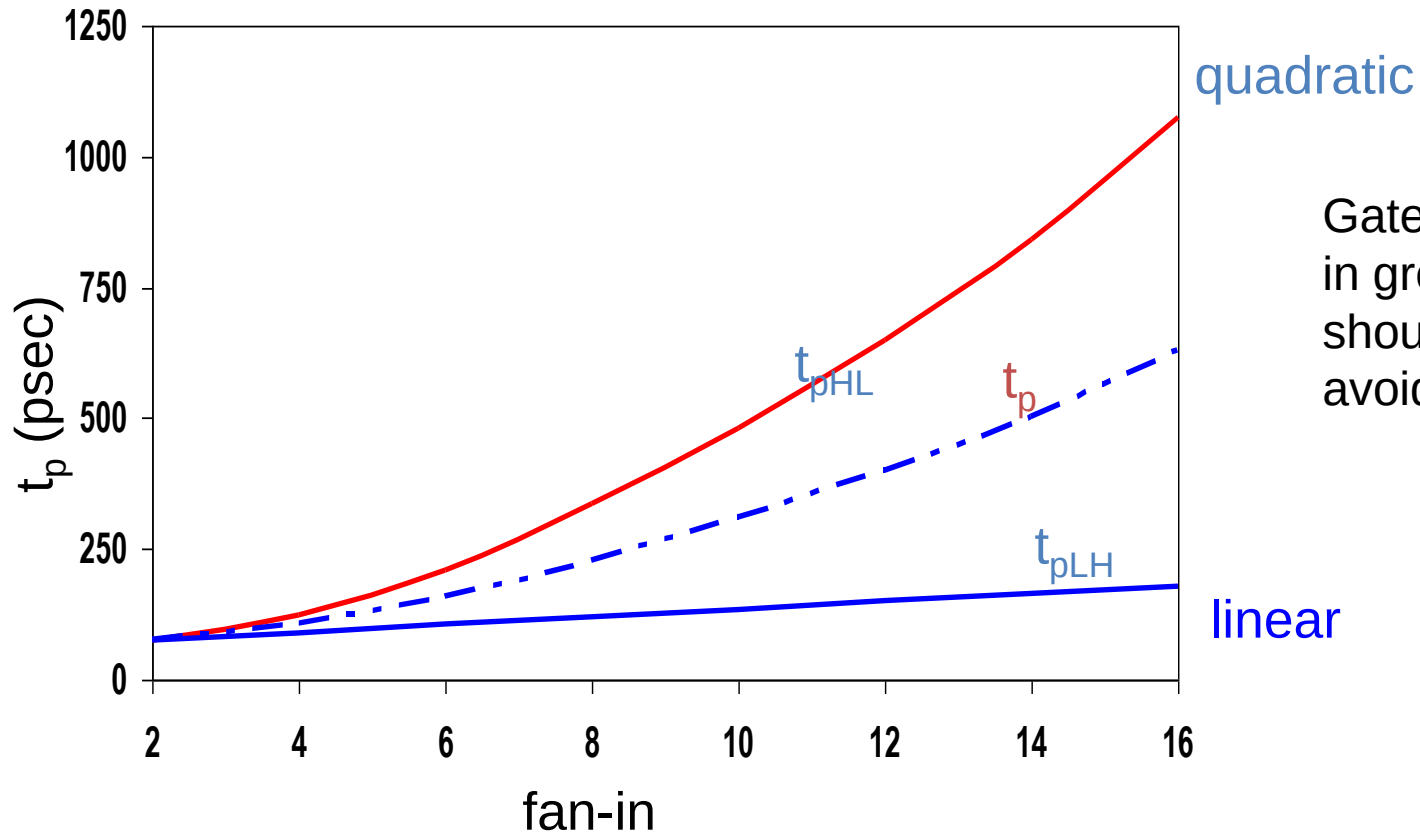


$$t_{pHL} = 0.69 R_{eqn}(C_1 + 2C_2 + 3C_3 + 4C_L)$$

Propagation delay deteriorates rapidly as a function of fan-in – **quadratically** in the worst case.



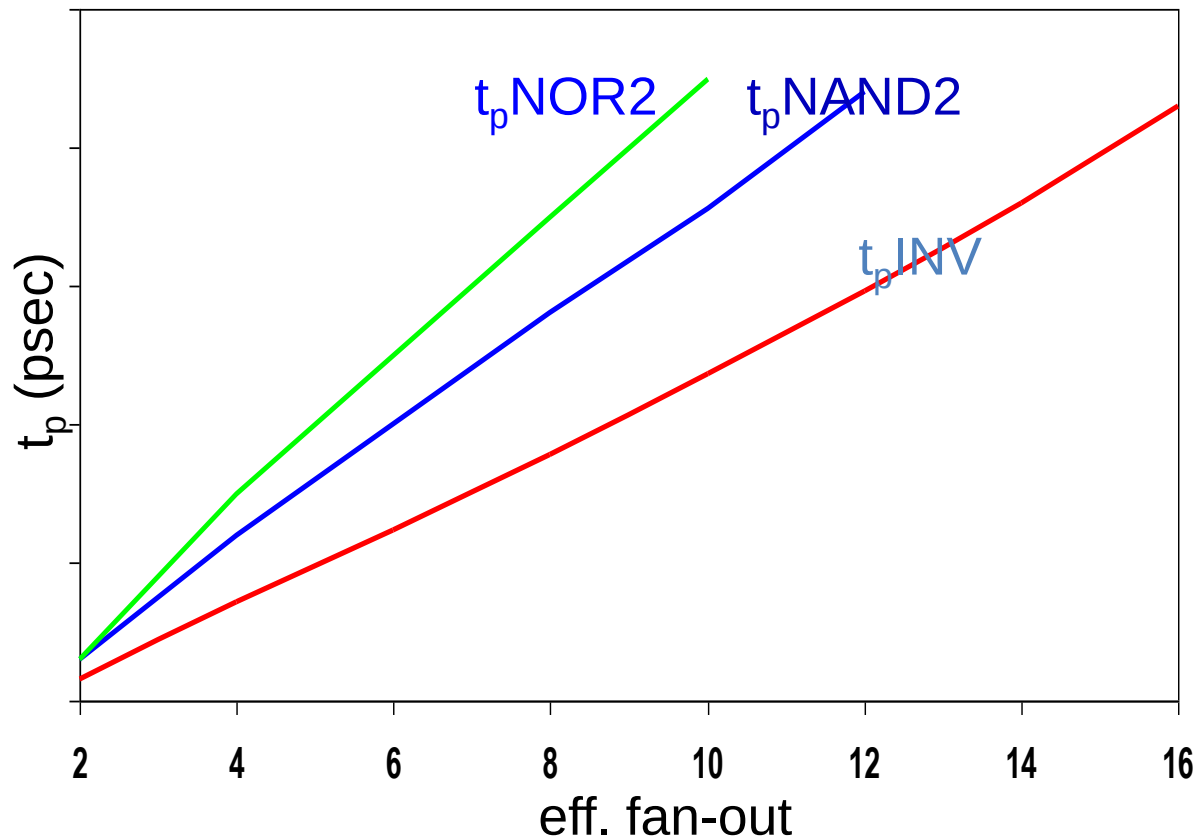
t_p as a Function of Fan-In



Gates with a fan-in greater than 4 should be avoided.

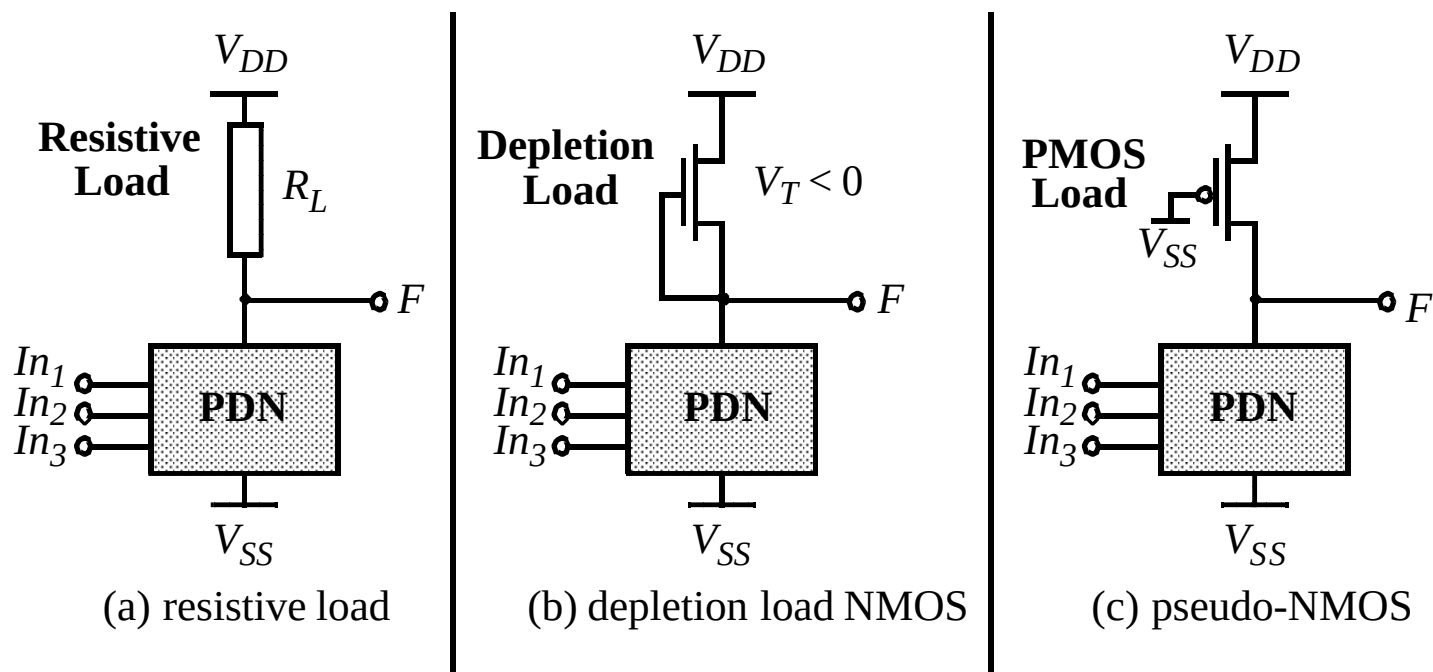


t_p as a Function of Fan-Out





Ratioed Logic

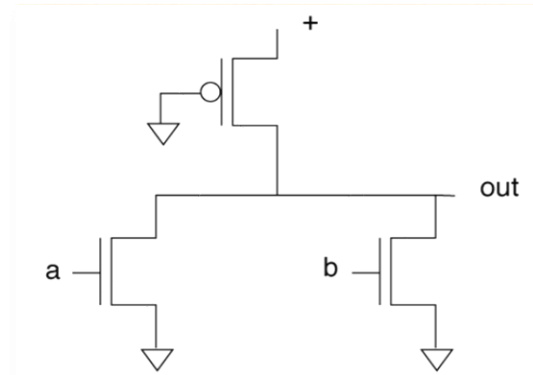


Goal: to reduce the number of devices over complementary CMOS



Pseudo-nMOS

- Uses a p-type as a resistive pullup, n-type network for pulldowns.



Consumes static power.

Has much smaller pullup network than static gate.

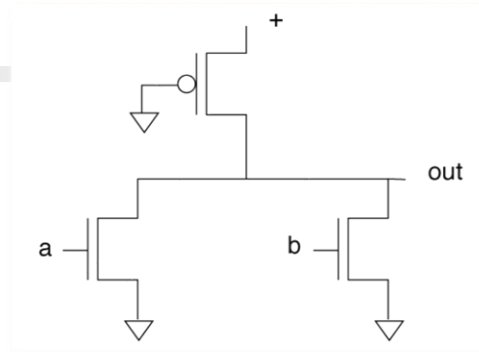
Pullup time is longer.

Wiring is simple.



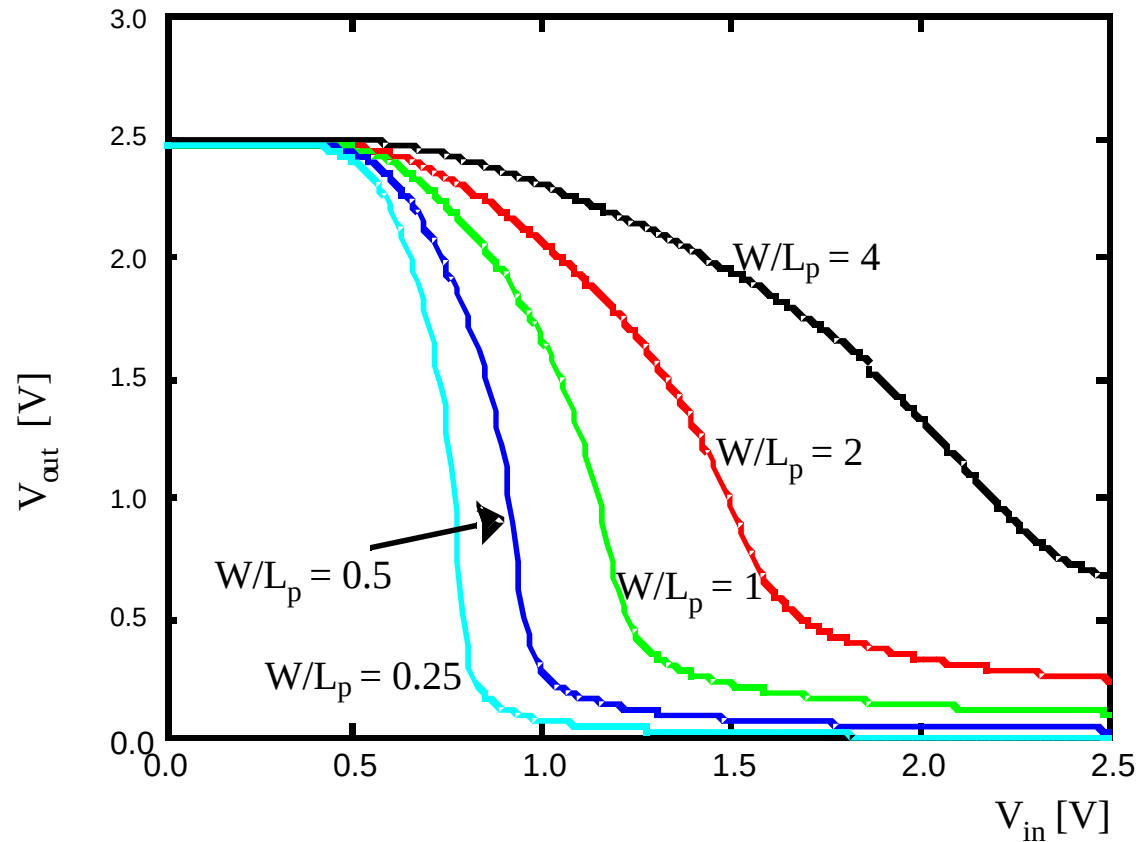
Producing output voltages

- Logic 1 output is always at V_{DD} .
- Logic 0 output is above V_{SS} .
- $V_{OL} = 0.2 (V_{DD} - V_{SS})$ is one plausible choice.
- For logic 0 output, pullup and pulldown form a voltage divider.
- Must choose n, p transistor sizes to create effective resistances of the required ratio.
- Effective resistance of pulldown network must be computed in worst case—series n-types means larger transistors.





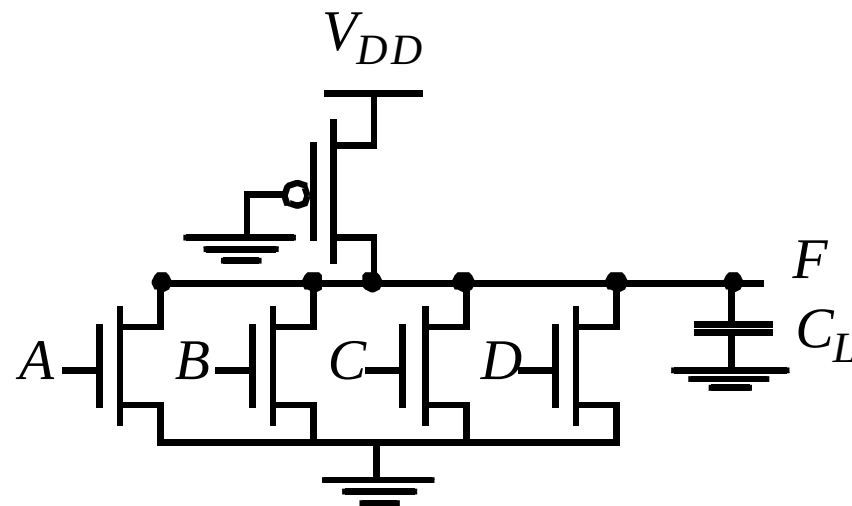
Pseudo-NMOS VTC





Pseudo-NMOS

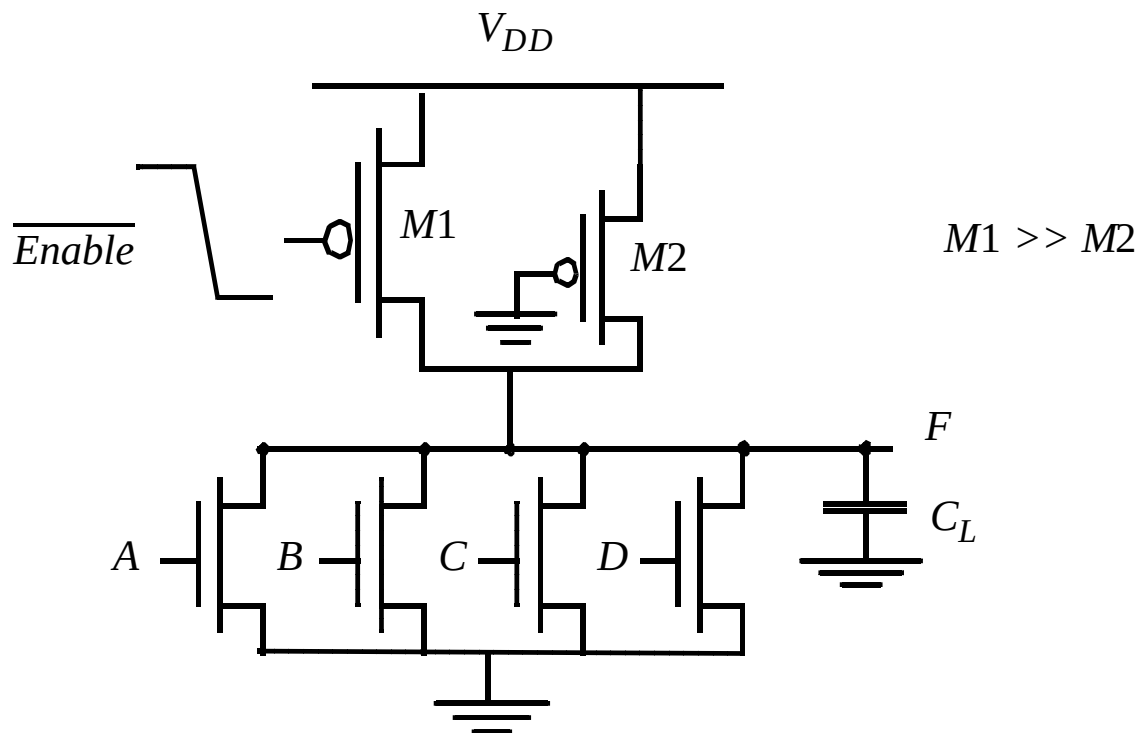
- Small area and load, but static power dissipation



$$V_{OH} = V_{DD} \text{ (similar to complementary CMOS)}$$



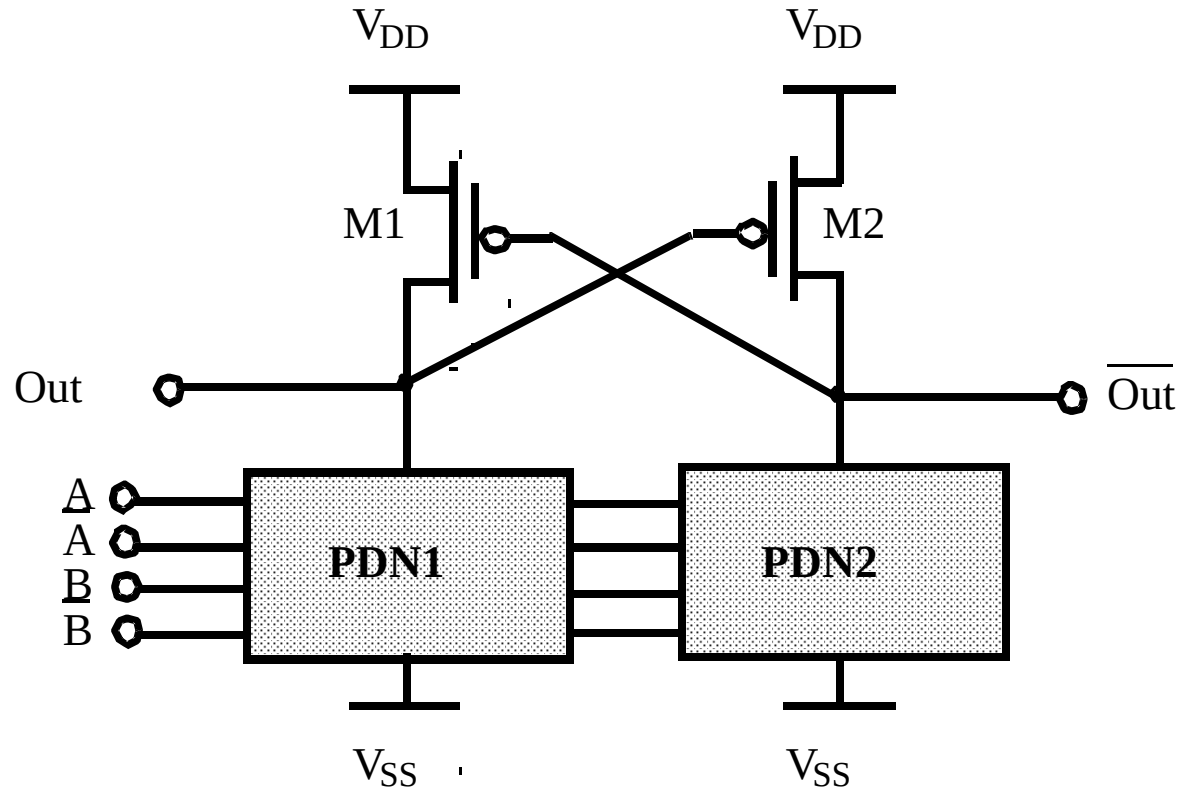
Improved Loads



Adaptive Load



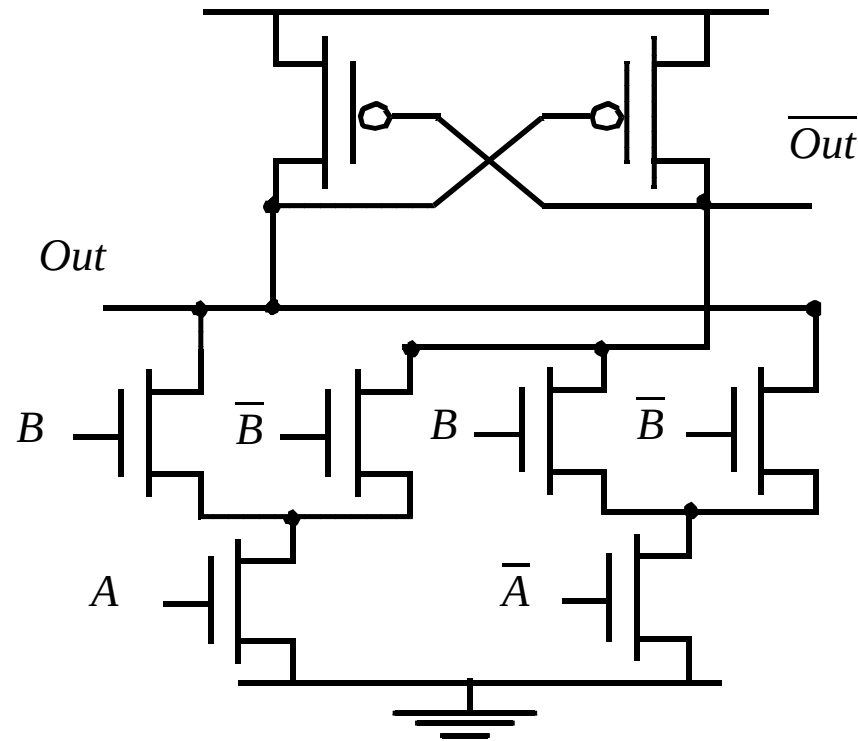
Improved Loads (2)



Differential Cascode Voltage Switch Logic (DCVSL)



DCVSL Example



XOR-NXOR gate

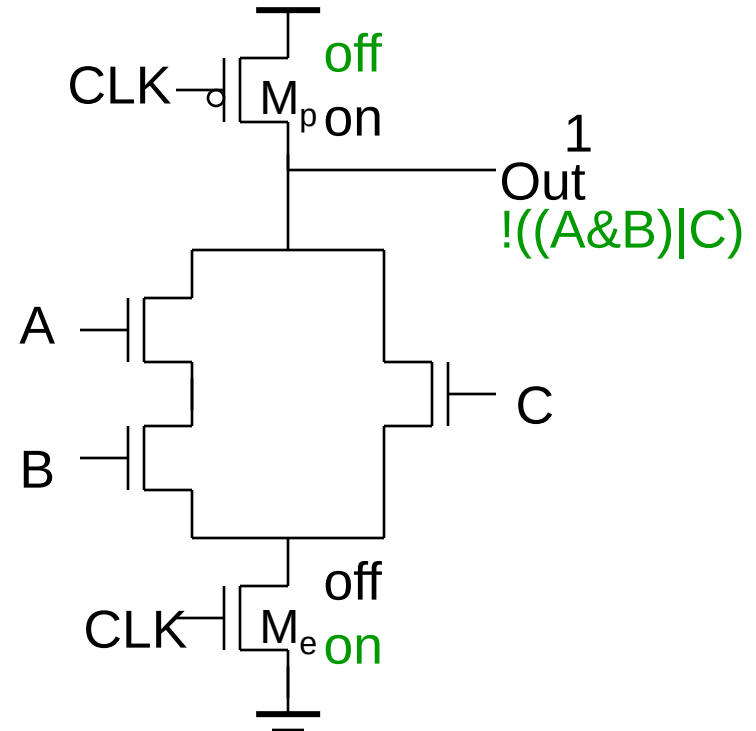
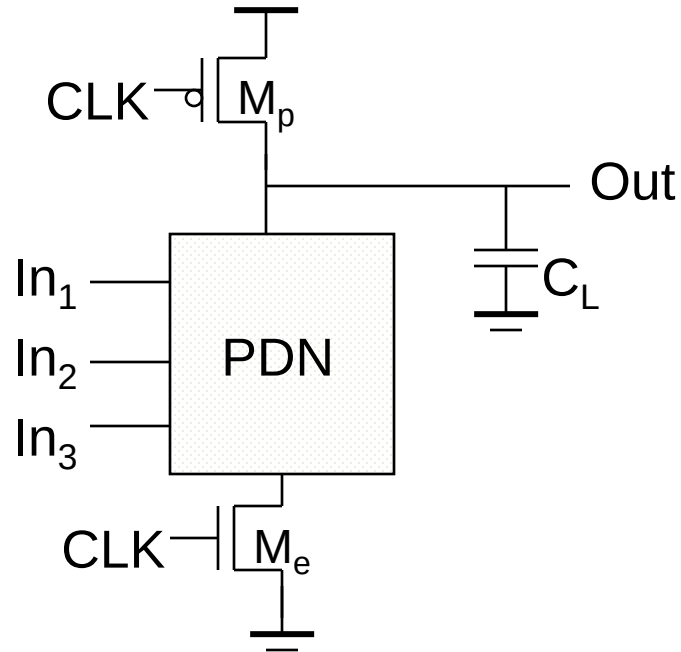


Dynamic CMOS

- In **static** circuits at every point in time (except when switching) the output is connected to either GND or V_{DD} via a low resistance path.
 - Complementary: fan-in of N requires 2N devices
- **Dynamic** circuits rely on the temporary storage of signal values on the capacitance of high impedance nodes.
 - requires only $N + 2$ transistors
 - takes a sequence of **precharge** and conditional **evaluation** phases to realize logic functions



Dynamic Gate



Two phase operation

Precharge (CLK = 0)

Evaluate (CLK = 1)



Conditions on Output

- Once the output of a dynamic gate is discharged, it cannot be charged again until the next precharge operation.
- Inputs to the gate can make **at most** one transition during evaluation.
- Output can be in the high impedance state during and after evaluation (PDN off), state is stored on C_L



Properties of Dynamic Gates

- Logic function is implemented by the PDN only
 - number of transistors is $N + 2$ (versus $2N$ for static complementary CMOS)
 - should be smaller in area than static complementary CMOS
- Full swing outputs ($V_{OL} = \text{GND}$ and $V_{OH} = V_{DD}$)
- Nonratioed - sizing of the devices is not important for proper functioning (only for performance)
- Faster switching speeds

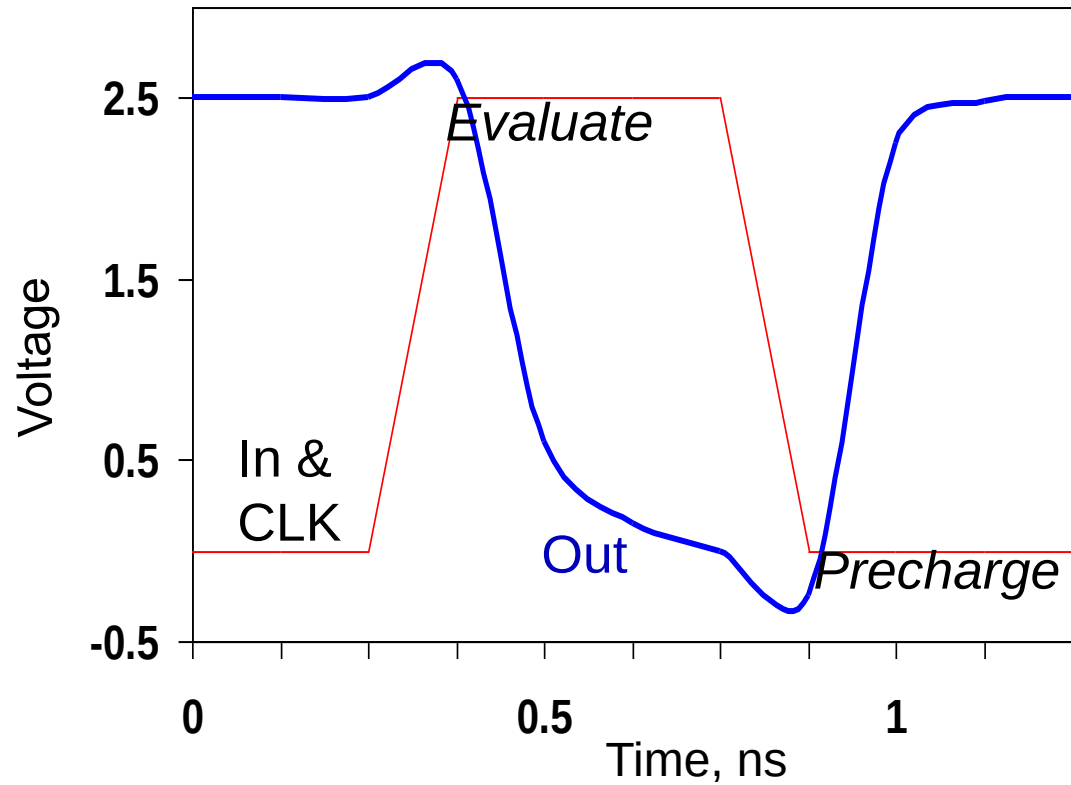
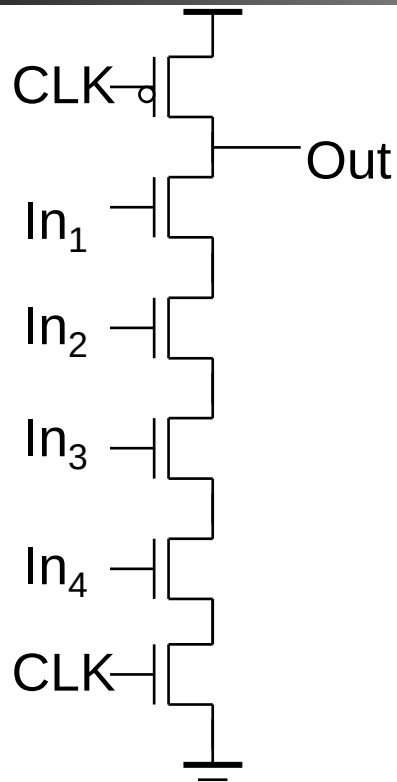


Properties of Dynamic Gates, con't

- Power dissipation should be better
 - consumes only dynamic power – no short circuit power consumption since the pull-up path is not on when evaluating
 - lower C_L - both C_{int} (since there are fewer transistors connected to the drain output) and C_{ext} (since there the output load is one per connected gate, not two)
 - by construction can have at most one transition per cycle – **no glitching**
- Needs a precharge clock

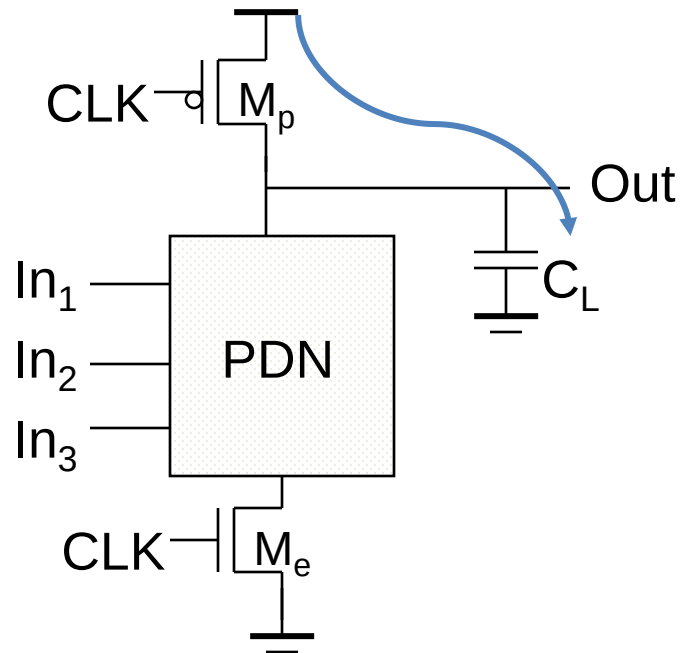


Dynamic Behavior





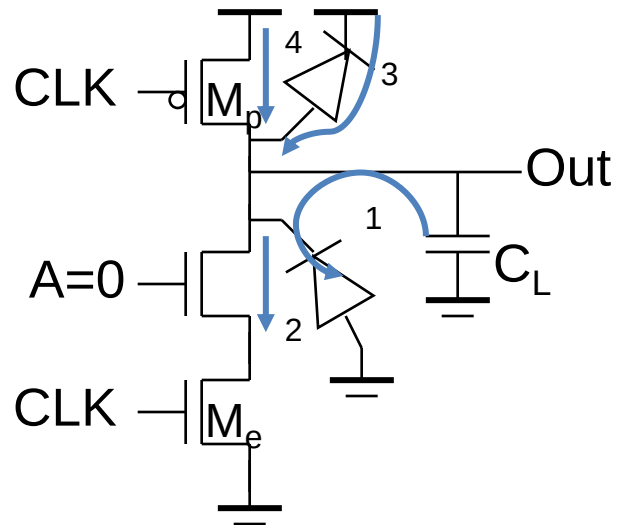
Power Consumption of Dynamic Gate



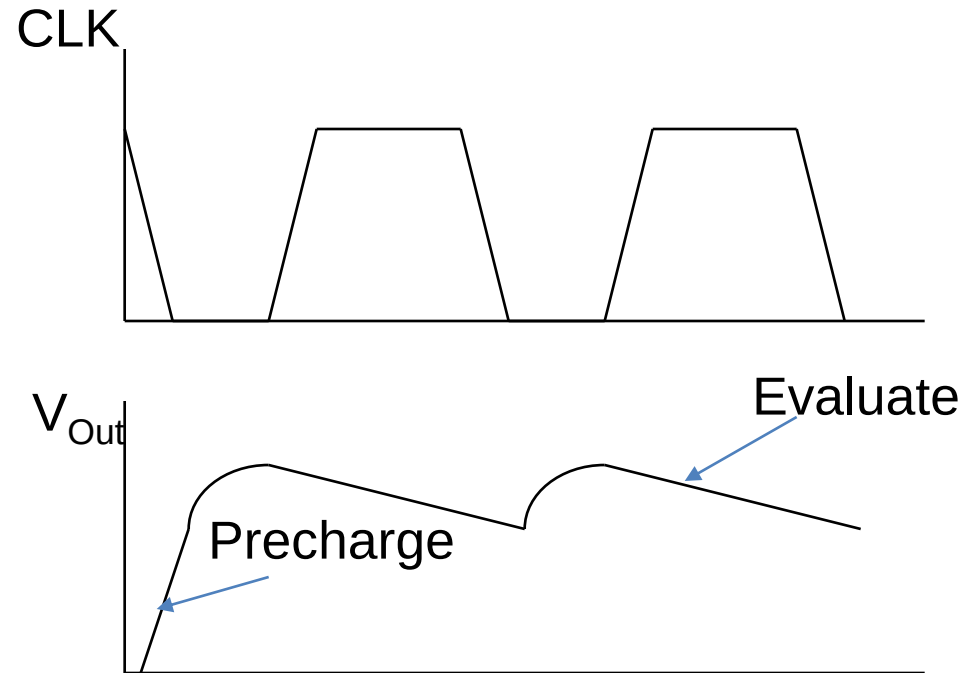
Power only dissipated when previous Out = 0



Issues in Dynamic Design 1: Charge Leakage



Leakage sources

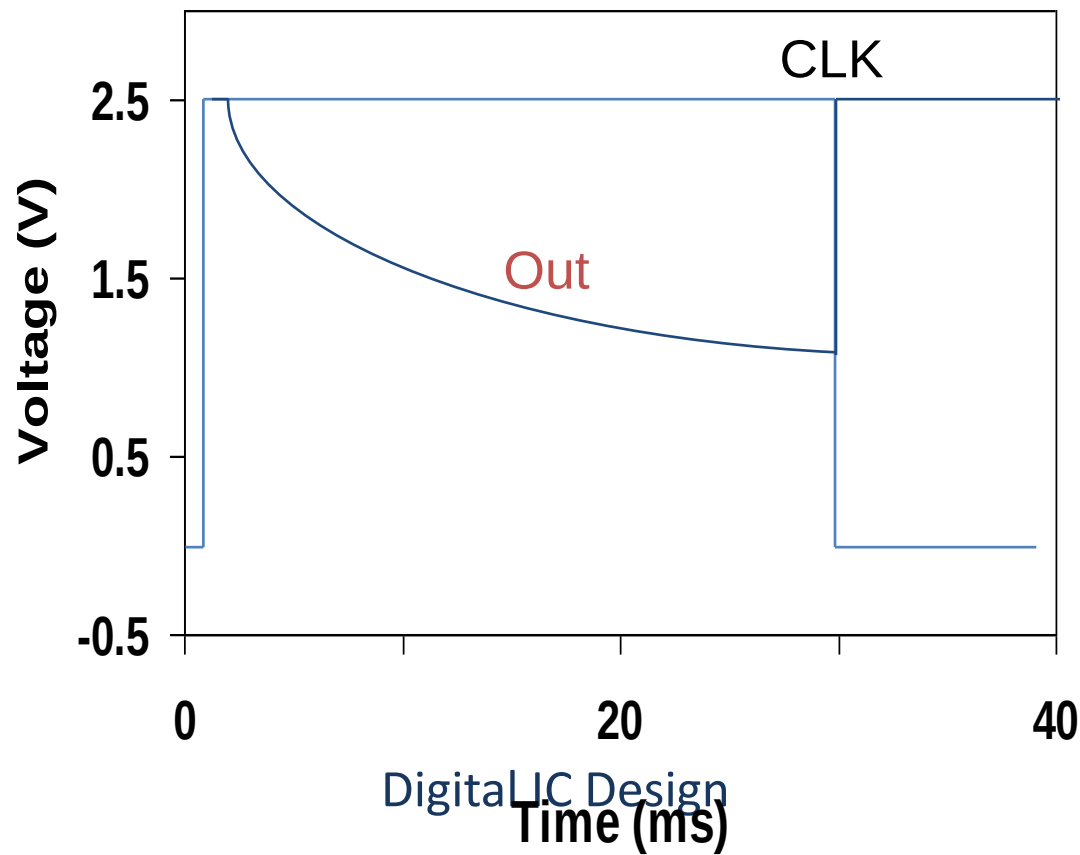


Minimum clock rate of a few kHz



Impact of Charge Leakage

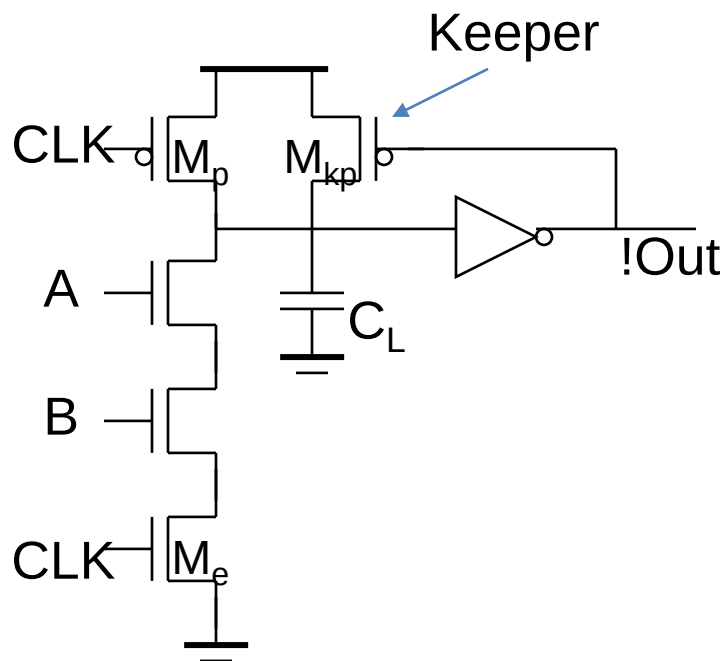
- Output settles to an intermediate voltage determined by a resistive divider of the pull-up and pull-down networks
 - Once the output drops below the switching threshold of the fan-out logic gate, the output is interpreted as a low voltage





A Solution to Charge Leakage

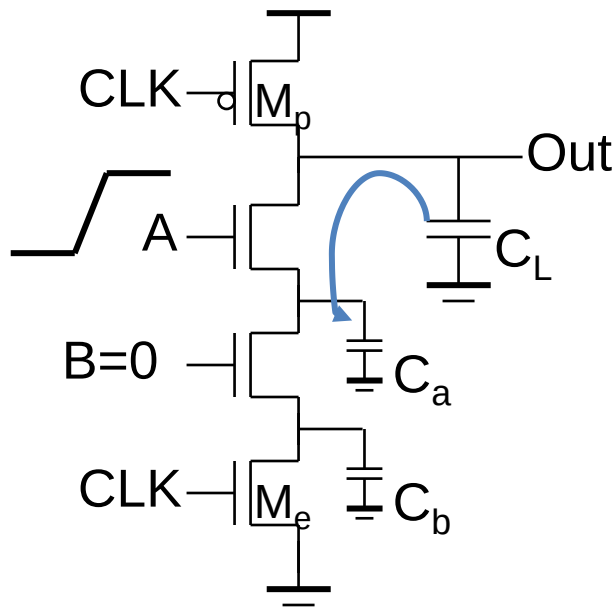
- Keeper compensates for the charge lost due to the pull-down leakage paths.



Same approach as level restorer for pass transistor logic



Issues in Dynamic Design 2: Charge Sharing

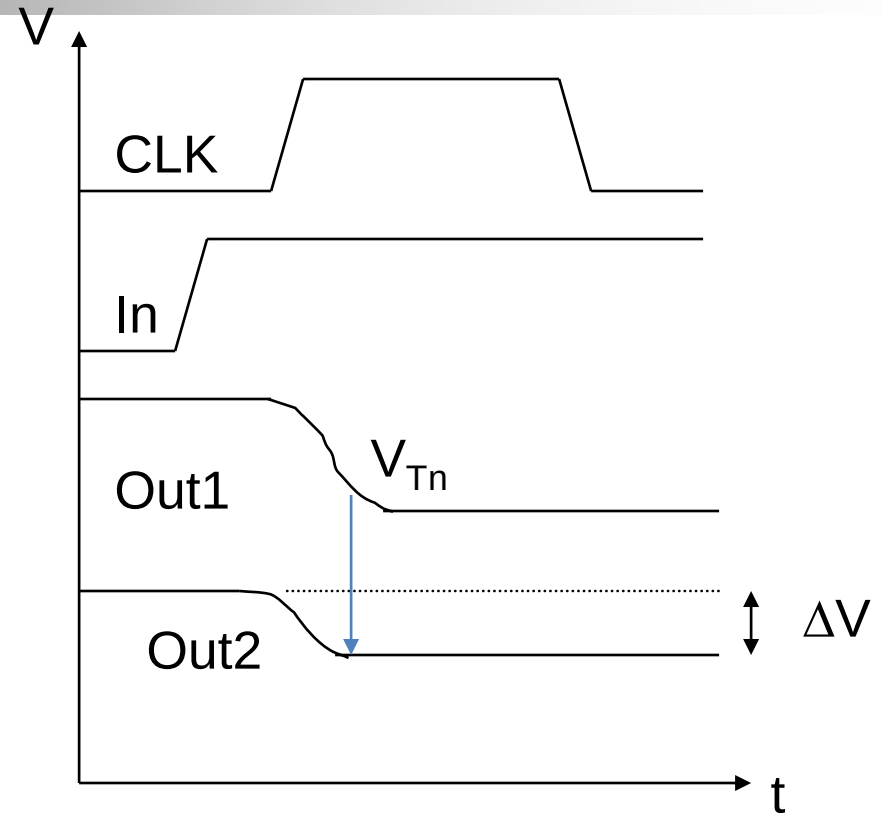
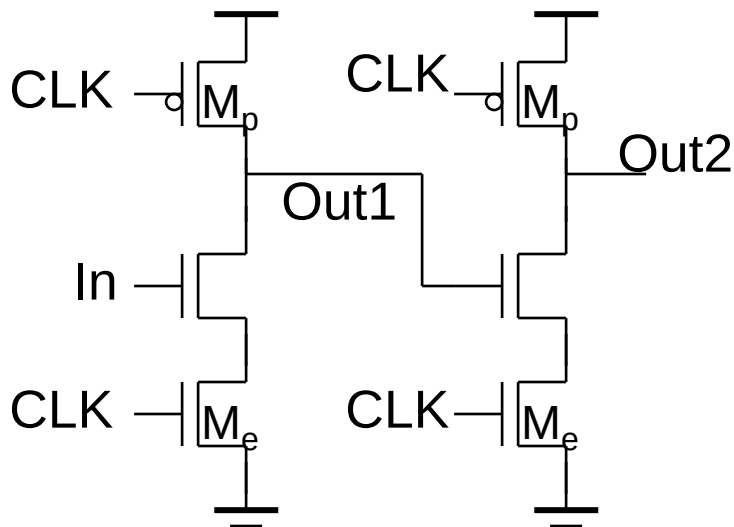


Charge stored originally on C_L is redistributed (shared) over C_L and C_A leading to static power consumption by downstream gates and possible circuit malfunction.

When $\Delta V_{out} = -V_{DD} (C_a / (C_a + C_L))$ the drop in V_{out} is large enough to be below the switching threshold of the gate it drives causing a malfunction.



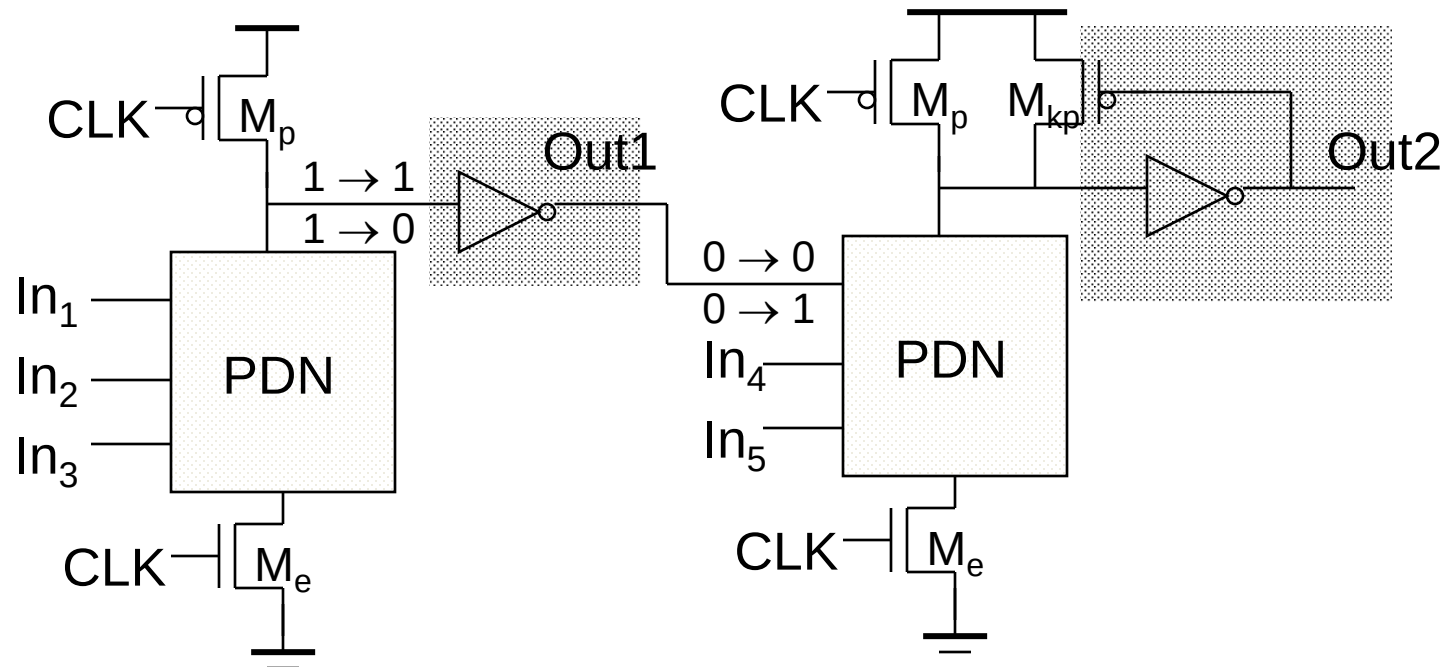
Cascading Dynamic Gates



Only a single 0 → 1 transition allowed at the inputs during the evaluation period!



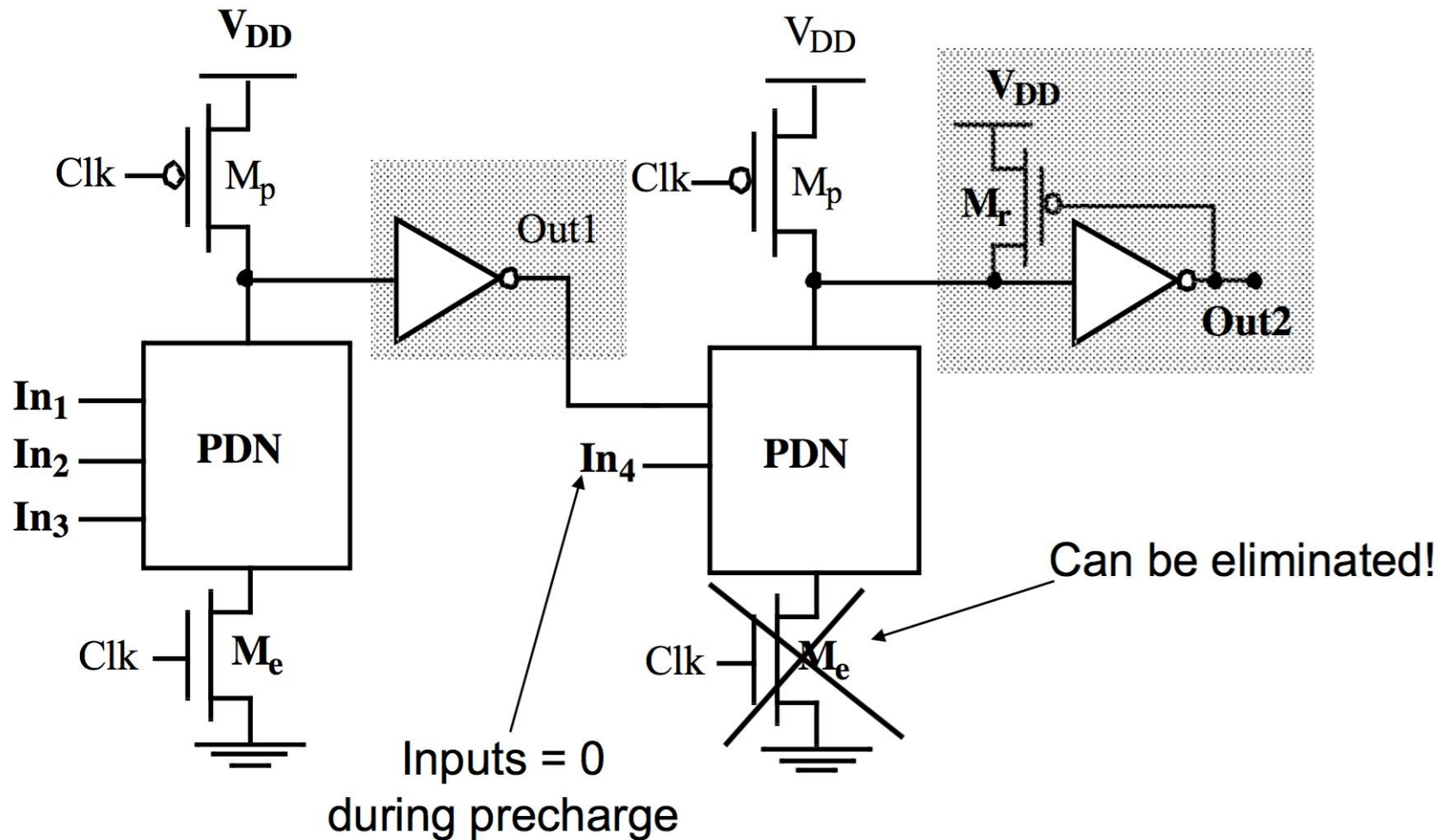
Domino Logic



- Only non-inverting logic can be implemented
- Very high speed
 - static inverter can be skewed, only L-H transition
 - Input capacitance reduced – smaller logical effort

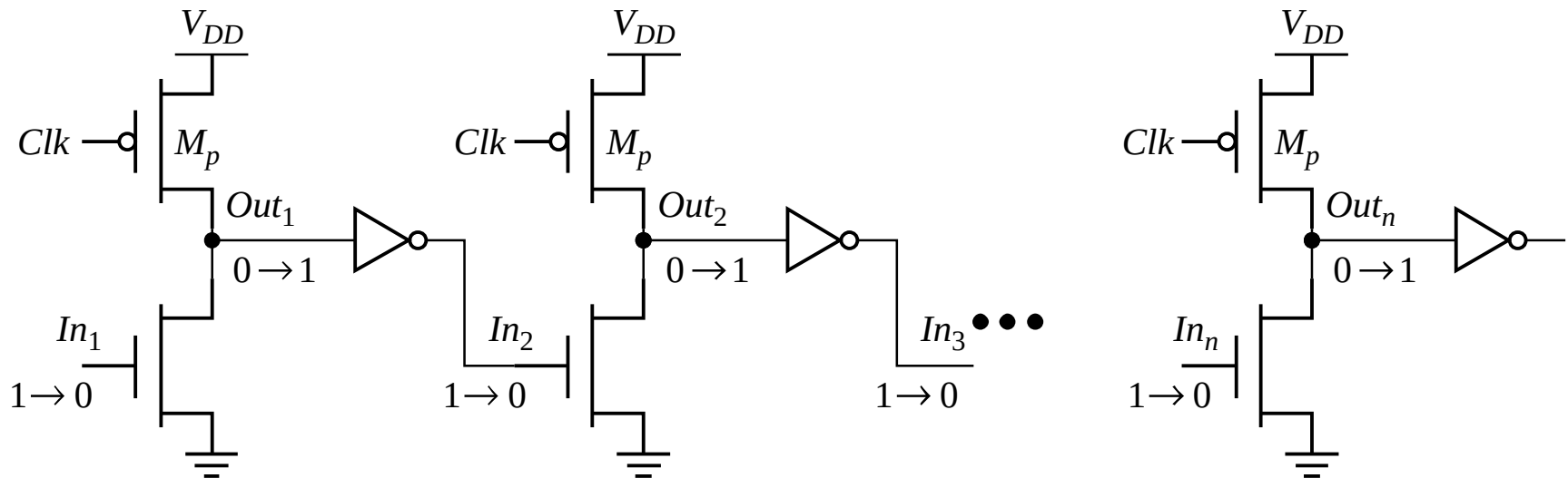


Designing with Domino Logic





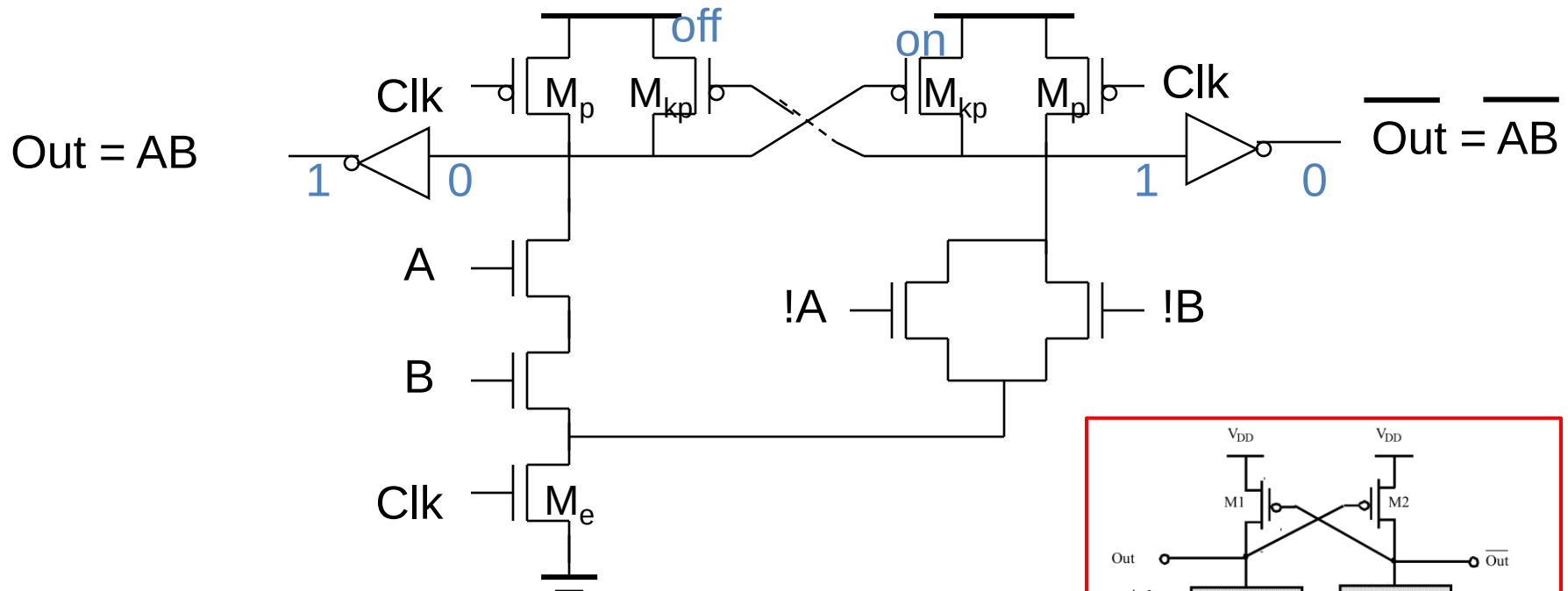
Footless Domino



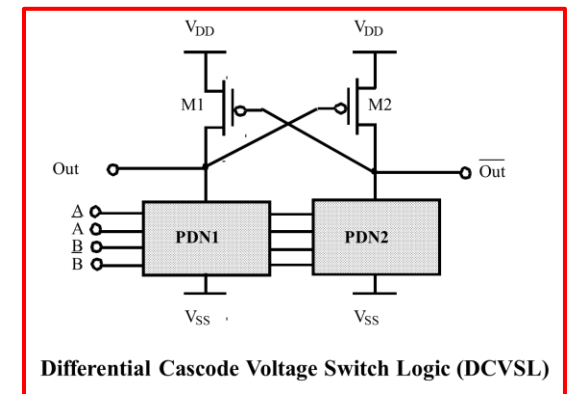
The first gate in the chain needs a foot switch
Precharge is rippling – short-circuit current
A solution is to delay the clock for each stage



Differential (Dual Rail) Domino



Solves the problem of non-inverting logic





Properties of Domino gates

- Logic function is implemented by the PDN only requires $N + 2$ transistors (versus $2N$ for static complementary CMOS)
- Should be smaller in area than static complementary CMOS
- Full swing outputs ($V_{OL} = GND$ and $V_{OH} = V_{DD}$)
- Faster switching speeds:
Input capacitance reduced
(since there are fewer transistors connected to the drain output) and C_{ext} (since there the output load is one per connected gate, not two)
- Inverter is required;
- Needs a precharge clock.

表 1 实现不同电路功能三种结构的
管子数量与延迟时间的比较

结构	2 输入与门	6 输入与门	6 - 64 译码器
动态多米诺逻辑	6/50ps	16/86ps	584/210ps
静态逻辑	6/60ps	16/120ps	744/260ps



How to Choose a Logic Style

- Must consider ease of design, robustness (noise immunity), area, speed, power, system clocking requirements, fan-out, functionality, ease of testing
- Current trend is towards an increased use of complementary static CMOS: design support through DA tools, robust, more amenable to voltage scaling.



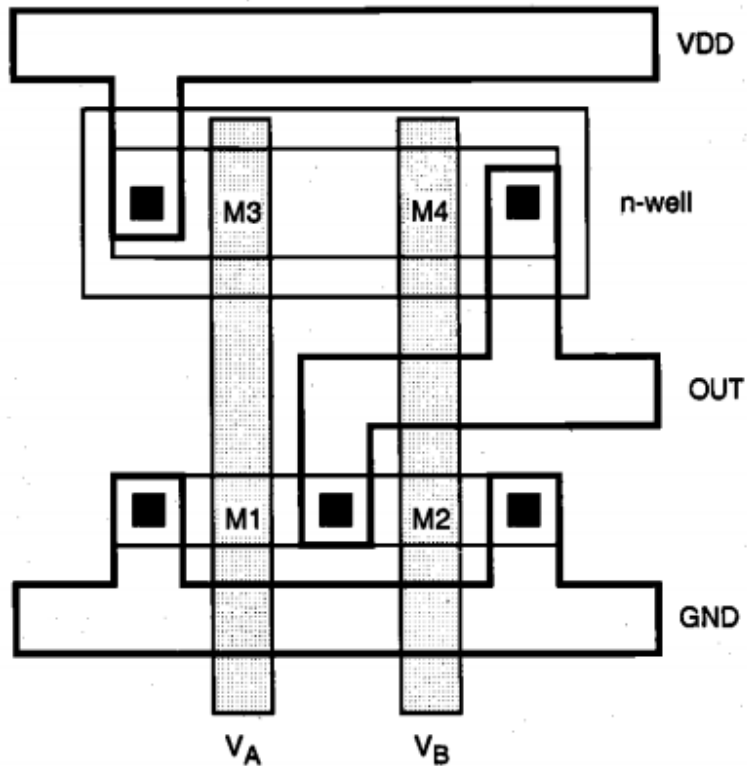
Summary

- Static complementary CMOS combines dual pull-down and pull-up networks. The performance of a CMOS gate is a strong function of the fan-in, and also is a function of the fan-out.
- The ratioed logic style consists of an active pull-down network connected to a load device.
- Dynamic logic: a two-phase scheme consisting of a precharge followed by an evaluation step.

Homework1

Question1-1: Circuit and Layout

Please draw the corresponding **transistor circuit diagram** of the layout below.



Homework1

Question1-2: logic gates

Sketch 5-input gates ($F = \overline{((AB + C)D + E)}$) using each of the following circuit techniques:

- a. Static CMOS
- b. Pseudo-NMOS
- c. Dynamic Gate

Answer the following question:

Which technique has the advantage of full swing?

Which one is not suitable for low frequency design?

Homework1

Question1-3:

What is the logic function of the circuits in Figure 1?

Size the devices so that the output resistance is the same as that of an inverter with an NMOS $W/L = 2$ and PMOS $W/L = 6$.

Which input pattern(s) would give the worst and best equivalent pull-up or pull-down resistance?

(Copy the circuit diagram to your exercise book. Some of the gate size is already given)

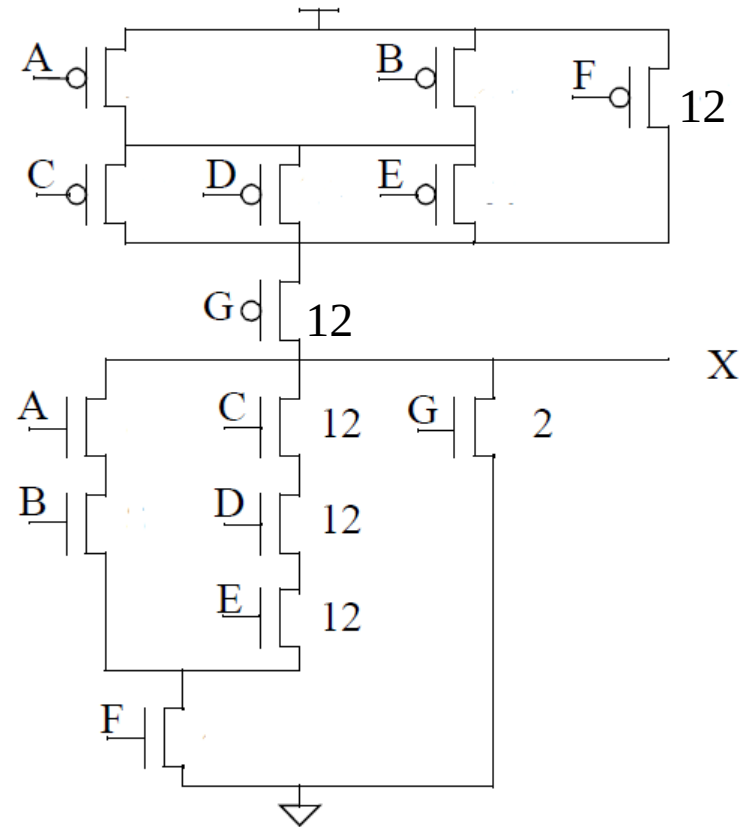


Figure 1

- Pls Send the electronic version of your homework to my DingTalk box (朱晓雷) **by Dec.31st,2024.**
THX!